BANK OF THAILAND

Monetary Policy Report

September 2015

Monetary Policy Report

The *Monetary Policy Report* is prepared quarterly by staff of the Bank of Thailand with the approval of the Monetary Policy Committee (MPC). It serves two purposes: (1) to communicate to the public the MPC's consideration and rationales for the conduct of monetary policy, and (2) to present the latest set of economic and inflation forecasts, based on which the monetary policy decisions were made.

The Monetary Policy Committee
September 2015

Mr. Prasarn Trairatvorakul	Chairman
Mrs. Pongpen Ruengvirayudh	Vice Chairman
Mr. Paiboon Kittisrikangwan	Member
Mr. Jamlong Atikul	Member
Mr. Porametee Vimolsiri	Member
Mr. Veerathai Santiprabhob	Member
Mr. Sethaput Suthiwart-Narueput	Member

Monetary Policy in Thailand

The Monetary Policy Committee

Under the Bank of Thailand Act, the Monetary Policy Committee (MPC) comprises the Governor and two deputy Governors, as well as four distinguished external members representing various sectors of the country, with the aim of ensuring that monetary policy decisions are effective and transparent.

The Monetary Policy Objective

The MPC sets monetary policy to promote the objective of supporting sustainable and full potential economic growth, without causing inflationary problems or economic and financial imbalances or bubbles.

The Monetary Policy Target

The Cabinet approved the annual average headline inflation target of 2.5 $\pm$ 1.5 percent as the monetary policy target for 2015, in place of the quarterly average core inflation target of 0.5 – 3.0 percent. The new inflation target was jointly proposed by the MPC and the Minister of Finance. In the event that headline inflation deviates from the target, the MPC shall explain the reasons behind the target breach to the public, together with measures taken and estimated time to bring inflation back to the target.

The Monetary Policy Instrument

The MPC utilizes the 1-day bilateral repurchase transaction rate as the policy interest rate to signal the monetary policy stance.

Evaluation of Economic Conditions and Forecasts

The Bank of Thailand takes into account information from all sources, the macroeconomic model, data from each economic sector, as well as surveys of large enterprises, together with small and medium-sized enterprises from all over the country, and various financial institutions to ensure that economic evaluations and forecasts are accurate and cover all aspects, both at the macro and micro levels.

Monetary Policy Communication

Recognizing the importance of monetary policy communication to the public, the MPC employs various channels of communication, both in Thai and English, such as (1) organizing a press statement at 14.00 on the day of the Committee meeting, (2) publishing edited minutes of the MPC meeting two weeks after the meeting, and (3) publishing the Monetary Policy Report every quarter.

Monetary Policy Report

September 2015

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Growth and Inflation Prospects and Monetary Policy

1. Growth and Inflation Prospects and Monetary Policy

The Thai economy is projected to expand by less than the previous assessment on the back of larger-than-expected exports contraction amidst a slowdown in trading partners' economies. Subdued farm prices and adverse impacts of drought dampened household confidence and spending, weakening the pace of economic recovery. Weak overall economic activities undermined private confidence and caused delays in private investment. Nevertheless, public spending and sustained tourism growth provided some support for the economy. The incident in Bangkok is, however, expected to reduce the number of foreign tourists in the near term.

Inflationary pressures fell mainly because of supply-side factors. Headline inflation is expected to remain negative throughout 2015, but it is likely to turn positive in the first quarter of 2016 as the effects of high base oil price in 2014 begin to wane. Core inflation declined slightly but is projected to remain positive throughout the forecast period.

The MPC maintained the policy rate at the August and September 2015 meetings, giving due weight to maintaining financial market stability amidst heightened volatility. Overall monetary conditions remained accommodative, while higher economic downside risks had already been, in part, taken into consideration, resulting in policy rate reductions during the first half of this year. Going forward, the MPC believes that monetary policy stance should remain accommodative to support economic recovery, and should help buoy market confidence as market sentiments remain sensitive to developments in the global financial markets. In addition, the MPC placed emphasis on evaluating the effectiveness of monetary policy transmission to the real economy, and continues to monitor the risk of a build-up of financial imbalances as a result of a prolonged low interest rate environment.

1.1 Growth and inflation prospects

The Thai economy is likely to recover at a slower pace than assessed in the previous issue of Monetary Policy Report. Slowdown in trading partners' economies, especially China and Asia, dented orders for Thai exports. Oil-related prices, such as petroleum products, declined in tandem with oil prices, causing exports contraction to be worse than formerly assessed. Mid-year drought affected farm outputs, particularly in-season rice. The incident in Bangkok caused a fall in the number of foreign tourists in the near term.

Weaker exports and drought impacts weighed down incomes and dented confidence amongst the household sector, causing them to be more cautious with spending. These meant that lower oil prices could not boost household consumption as effectively. Moreover, weak domestic and external demands prompted the business sector to postpone investment plans, particularly in machinery and equipment for production capacity expansion, although some private investments, with links to government projects, have been made.

Public spending exceeded the previous projection and provided support to the economy, especially with contracts being expedited for government investment projects. The new Cabinet's economic stimulus also focused on boosting investment and employment.

Nevertheless, higher public spending has been unable to counter the impacts of weaker exports and private spending. **The MPC thus revises down the GDP growth forecasts to 2.7**

percent in 2015 and 3.7 percent in 2016 (see Table 1.1).

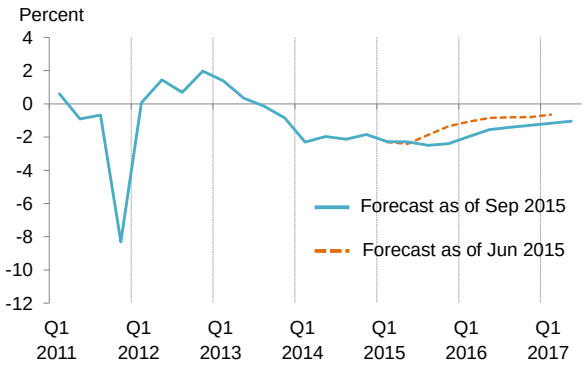
Inflationary pressures turned out to be lower than the assessment in the last Monetary Policy Report, due mainly to the supply-side factors. The Committee hence revised down headline and core inflation forecasts throughout the projection period. Headline inflation forecast moderated due to cost factors, namely falling global crude oil prices. Meanwhile, raw food prices also remained low. Therefore, the MPC expected headline inflation to remain negative for an extended period, before gradually edging up and turning positive in 2016 Q1 when the effects of high base oil price in 2014 disappear. Production costs declined in line with oil prices, and demand-side pressures fell. This reflects the slower closing of output gap, consistent with weaker-than-projected economic recovery. Core inflation is thus projected to decline in 2016 **(Table 1.1 and Chart 1.1)**. However, the Committee assessed deflationary risks to be low, as the negative headline inflation was primarily caused by supply-side factors, especially plunging oil prices. Meanwhile, core inflation remains positive throughout the forecast period, while inflation expectations remained close to the inflation target.

Table 1.1 Forecast summary

Percent	2014*	2015	2016
GDP growth	0.9	2.7	3.7
		(3.0)	(4.1)
Headline inflation	1.9	-0.9	1.2
		(-0.5)	(1.6)
Core inflation	1.6	1.0	0.8
		(1.0)	(1.0)

Note: * Outturn
( ) MPR June 2015
Source: Office of National Economic and Social Development Board, Ministry of Commerce and calculations by Bank of Thailand

Chart 1.1 Output Gap



The MPC revised down the projections for the headline inflation and core inflation -0.9 and 1.0 percent in 2015 and 1.2 and 0.8 in 2016, respectively.

The Committee factored the following developments into the growth and inflation forecasts.

(1) Trading partners' economies are set to expand at a slower rate than previously

assessed, mainly due to the slowdown in China and Asia (Table 1.2).

The Chinese economy is assessed to grow at a lower rate than the previous projection, resulting in lower growth in other Asian economies. Given that China is a major importer of commodities and electronic goods, the Chinese slowdown lowered prices of commodities, especially agricultural raw materials and metals, and prices of electronics. Furthermore, given close linkages between China and other Asian economies, the Chinese slowdown and the renminbi devaluation caused Asian currencies to weaken (*Box 1: Impacts of the Chinese Economic Slowdown and the renminbi Devaluation on Thai Exports*).

The slowdown in the Chinese and Asian economies caused a slowdown in Thai exports, especially exports of final products to China. However, Thai exports of intermediate goods to China benefited somewhat from stronger Chinese exports boosted by the weaker renminbi. Although some Thai exports are classified to be in the same categories as Chinese exports, the renminbi devaluation is likely to have limited impacts on Thai exports to other countries. This is because, upon closer examination, Thailand and China export different types of goods, even if they are listed to be under the same category, and do not compete for the same markets.

Weaker baht vis-à-vis the U.S. dollar had provided limited boost to orders for Thai exports, due to sluggish economic conditions in major export destinations and depreciation of the currencies of other export competitors. The weaker baht also led to lower prices in certain export products as a result of price negotiation, especially

Chart 1.2 Growth assumptions for Thailand's trading partners

Annual percentage change %YoY	Weight (%)	2014	2015		2016	
			Jun 2015	Sep 2015	Jun 2016	Sep 2016
The U.S.	14.9	2.4	2.1	2.5	2.9	2.8
The euro area	10.0	0.9	1.3	1.3	1.6	1.5
Japan	13.6	-0.1	0.8	0.8	1.4	1.2
China	15.7	7.4	6.9	6.8	6.8	6.6
Asia (excluding Japan and China)*	37.4	4.2	4.0	3.7	4.4	4.2
Total**	100	3.5	3.3	3.3	3.8	3.6

Note: * Weighted by each trading partner's share in Thailand's total exports in 2014 (7 countries: Singapore (6.5%), Hong Kong (7.9%), Malaysia (8.0%), Taiwan (2.5%), Indonesia (5.9%), Republic of Korea (2.8%) and the Philippines (3.7%))
**Weighted by each trading partner's share in Thailand's total exports in 2014 (13 countries)

those products in which Thai exporters had little bargaining power.

Moreover, volatility in the global financial market rose in 2015 Q3, in part as a result of weaker-than-expected Chinese growth, fragile global economic recovery as well as low inflationary pressures in the U.S. in line with stronger dollar and falling commodity prices. Thus, the MPC were of the view that, despite recent positive developments in the U.S., the Fed is likely to postpone policy rate increase until late-2015, instead of in 2015 Q3 as previously anticipated, in order to ensure steady recovery of the U.S. economy going forward.

Exports of services are likely to register steady growth with the potential to expand at a faster pace than previously assessed. Based on the upward trend of the number of foreign tourists prior to the bomb incident in Bangkok, the assumption on the number of foreign tourists in 2015 was raised from 28.8 to 29.4 million people. Although the incident may have affected confidence of some tourists, the MPC judged that this setback would only be short-lived and expected those who had cancelled their trips to Thailand to return provided that there is no further negative incident. Under this baseline projection of improvement in overall atmosphere and tourists' confidence, exports of services are thus projected to continue the expansion trend in 2016.

The further contraction of exports of goods adversely affected employment and income of households in manufacturing industry for exports. The declining export prices, especially rubber, weighed down incomes for farm households and resulted in more cautious household spending behavior, hindering economic recovery. This in

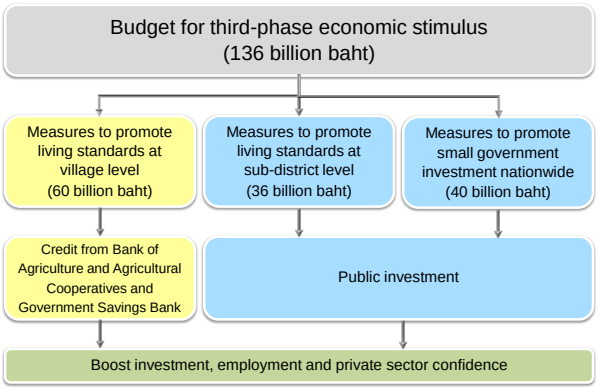
turn dented private sector confidence and caused postponements in private investment, especially in machinery and equipment for capacity expansion.

The Committee assessed that there is downside risk for trading partners' economies to expand at a slower pace than the baseline assumptions, due to weaker-than-projected Chinese growth as well as the adverse impacts of the Chinese slowdown and the renminbi devaluation on Asian economies. Furthermore, the Committee deems important to monitor the progress of economic and fiscal reforms in Greece, which continues to be hindered by country's weak economic fundamentals, as well as its long-term implications for the euro area.

(2) The public sector continued to provide support to the economy through government spending, especially in public investment.

The MPC revised upwards the assumptions on public spending. Central government spending increased as the government expedited the signing of contracts for public investment in 2015 Q3. Disbursements of extra-budgetary spending under the second phase of the fiscal stimulus were also processed faster than expected. Furthermore, additional investment projects were scheduled to commence in late-2015, namely motorway construction and phase III fiscal stimulus[1/] **(Chart 1.2)**. State enterprise investment was also boosted

Chart 1.2 Third-phase economic stimulus



Source: Cabinet's decision, September 1, 2015

[1/] Measures include: (1) measures to promote living standards at sub-district level, (2) measures to promote small government investment nationwide, as part of additional public investment spending in late-2015, and (3) measures to promote living standards at village level in the form of financial assistance to the private sector.

by public utility projects under the Electricity Generating Authority of Thailand and the Provincial Electricity Authority. Nonetheless, some projects faced delayed, such as the Green Line Mass Rapid Transit Project (MRT).

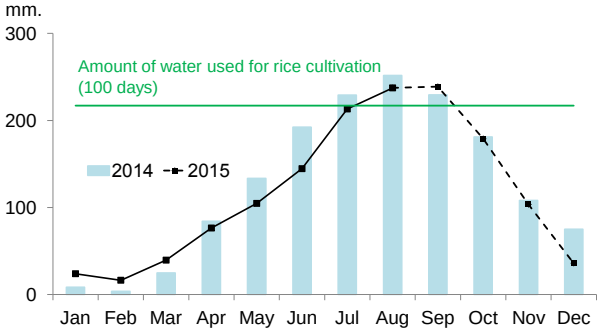
Looking ahead, the MPC agreed that sustained public spending should continue to underpin private sector confidence and boost private investment by firms involved with government projects, such as investment under the public-private partnership scheme for selected MRT projects. Moreover, the measure aiming at providing financial support at the village level under phase III fiscal stimulus should provide additional boost to small-scale investment and household consumption in the rural area.

The Committee assessed the upside risk of public spending being greater than the baseline assumption to be close to the downside risk of public spending being below it. The key upside risk factor is that the government may be able to accelerate spending of phase II and III stimulus measures faster than expected disbursement rate. The downside risk is that the new procurement system could cause delays in some disbursements of funds.

(3) Drought resulting from unusually low levels of rainfall during mid-2015 (Chart 1.3).

Unusually low amount of rainfall in mid-2015 combined with droughts that had lasted from the previous year led to water shortage for farming. The problem was particularly severe for in-season rice cultivation, which is mainly dependent on rainwater, causing a decline in in-season rice production in 2015. However, some farmers could resume cultivation, following the seasonal return of rainfalls in August 2015. Rice harvests were thus

Chart 1.3 Ratio of amount of rainfall to amount of water used for rice cultivation



Source: Thai Meteorological Department, Department of Irrigation, GISTDA, calculations by Bank of Thailand

postponed until early 2016, which would boost in-season rice production in 2016. The increase in in-season rice production next year should also compensate for the likely decline in off-season rice production stemming from low water levels in dams following this year's drought.

Drought and depressed farm prices reduced farm income, inducing some workers to migrate to seek jobs in non-farm sectors. However, most of them moved to jobs in the services sector, which offer relatively lower hourly wages and shorter working hours, including jobs in trading and construction sector. As a result, it led to a fall in average income level in the non-farm sector. **(Charts 1.4 and 1.5)**.

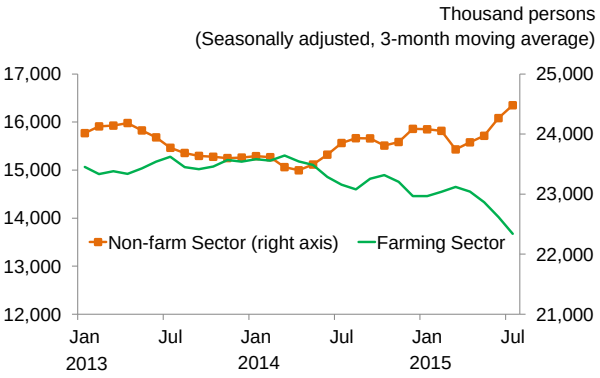
As for the impacts of drought on the price of rice, the decline in in-season rice production in 2015 did not lead to a significant increase in the price of Thai rice, owing to sizeable government stocks of rice and sluggish world demand.

Overall, drought reduced farm incomes and increased financial vulnerability of farm households. This group of households has already suffered from weak financial position because of their larger debt burden relative to other groups **(Chart 1.6)**. Hence, farm households are likely to exercise greater restraint in their spending in the latter half of 2015, likely resulting in the slowing of household spending growth.

(4) Global oil prices declined by greater amounts than expected in 2015 Q3 (Chart 1.7).

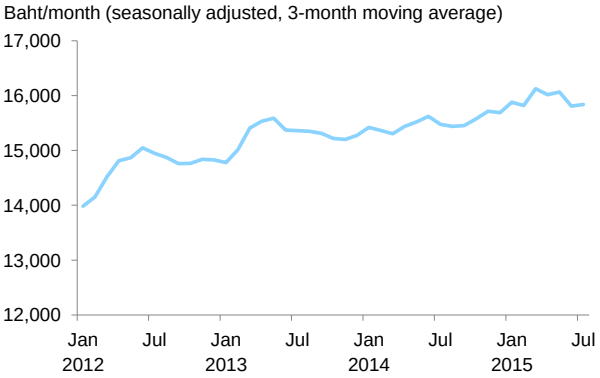
The MPC revised downwards the baseline assumptions for crude oil price throughout the forecast period, averaging 52.8 and 51.3 US dollar per barrel in 2015 and 2016, respectively. The revision followed the substantial decrease in oil

Chart 1.4 Employment



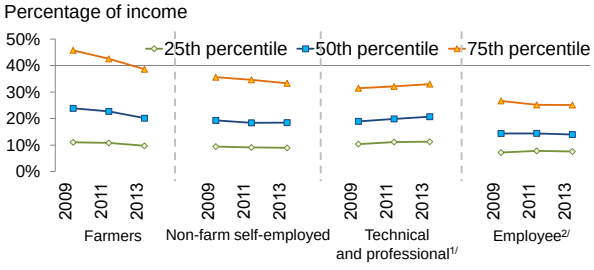
Source: National Statistical Office, calculations by the Bank of Thailand

Chart 1.5 Real average non-farm income



Source: National Statistical Office, calculations by the Bank of Thailand

Chart 1.6 Ratio of debt burden to household income, by occupation



Note: Only for households with debt burden
1/ Households employed in technical and professional works, such as manager, academia, professionals, technicians, etc.
2/ Households with employees, such as those employed in farms and fishery, machine operators, clerks, servers, skilled workers in production or construction works, etc.
Source: Household Economic and Social Survey Project, National Statistical Office, calculations by the Bank of Thailand

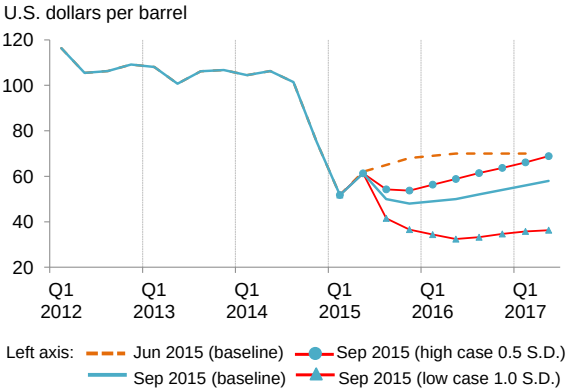
prices in 2015 Q3, on the back of higher global oil supply following increased oil production in the U.S., where some producers had resumed their operations, as well as high levels of production by OPEC members. Going forward, the lifting of sanctions on Iran should further increase global oil supply, while demand for oil continues to remain subdued due to fragile global recovery and the Chinese slowdown.

Declining global oil prices translated into falling domestic retail oil prices, lowering production and transport costs for the business sector. This led to a more prolonged period of negative headline inflation than previously assessed. Headline inflation is projected to turn positive in 2016 Q1, on the assumption of a gradual rise in global oil prices and the disappearance of high base oil price effects.

The decline in oil prices had not provided a significant boost to private consumption and investment because of other negative factors. Household income remained subdued due to depressed farm prices and drought effects. Meanwhile, household debt remained at high levels, discouraging household spending. The business sector had little need to invest because of considerable excess production capacity and weak-demands from both domestic and foreign markets.

Moreover, lower oil prices weighed down on the prices of commodity exports, causing a fall in the value of Thai exports and denting Thai exporters' income. Nevertheless, subdued oil prices also reduced the value of imports quite substantially. Thailand's current account thus continued to record a sizeable surplus.

Chart 1.7 Assumptions on Dubai oil price

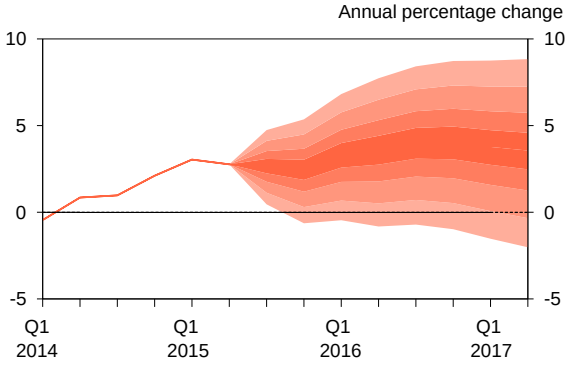


According to the MPC's evaluation, the risk that the crude oil price will be below the baseline assumption is higher than the risk that it will exceed the baseline assumption, due to the following factors: (1) Global demand could contract more than the current assessment. (2) Global supply could rise by more than previously assessed if OPEC member countries (especially Saudi Arabia and Iraq) decide to ramp up production to maintain market share. (3) Shale oil producers could expand production earlier than expected. On the other hand, crude oil price could rise above baseline assumption if the Middle East conflict escalates and causes a fall in global oil supply.

Risks to Growth and Inflation Forecasts

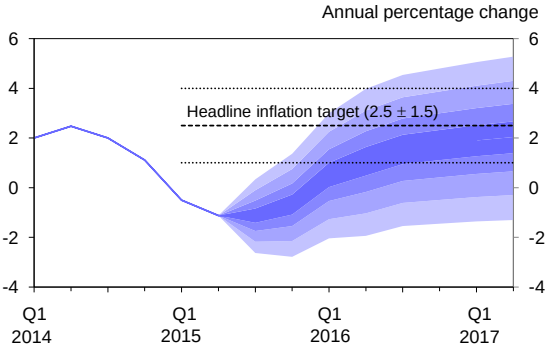
According to the MPC's assessment, **the likelihood that growth will be below the baseline projection is higher than the probability that it will be above the baseline projection. This assessment is depicted in the growth fan chart, which is skewed downwards throughout the forecast period (Chart 1.8).**

Chart 1.8 GDP growth forecast



Note: The fan chart covers 90 percent of the probability distribution.

Chart 1.9 Headline inflation forecast



Note: The fan chart covers 90 percent of the probability distribution.

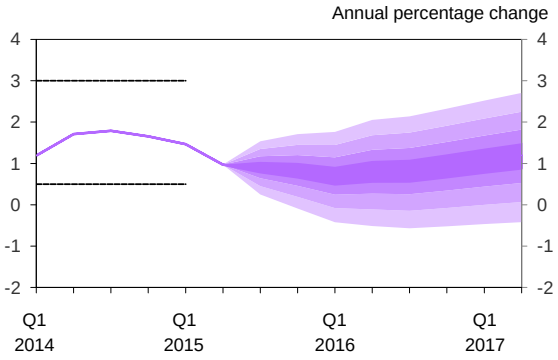
Downside risks to economic growth stem from the following: (1) Trading partners' economies, especially China, could grow more slowly than anticipated. (2) The adverse impact of the Chinese slowdown on Asian economies could be larger than expected. (3) Incident in Bangkok could result in fewer-than-expected number of tourists. (4) Public investment may not spur private investment as much as forecasted amidst low private sector confidence.

Upside risk factors to growth could arise from the following sources: (1) The number of tourists in 2016 could surpass the forecast.

(2) The phase II and phase III fiscal stimulus, and immediate financial support measures for SMEs,[2/] could be implemented quickly and result in more pronounced impacts than expected.

The Committee judges that **headline and core inflation are more likely to tread below the central projection than to surpass it. The greater downside risk is reflected in the inflation fan charts that are skewed downwards throughout the forecast period (Charts 1.9 and 1.10).** The assessment is based on the risk factors concerning crude oil prices and softer demand-side pressures as economic growth could fall short of the baseline projection.

Chart 1.10 Core inflation forecast



Note: The fan chart covers 90 percent of the probability distribution.

1.2 Monetary policy decision

The MPC maintained the policy rate, giving due weight to maintaining financial market stability amidst heightened global market volatility. Prior policy easing contributed to more accommodative monetary conditions. At present, overall monetary policy conditions remained accommodative. This was reflected by the exchange rate movements that stays conductive to the economy economic recovery and firms' cost of funds in the money market that had declined following the previous policy rate reductions.

[2/] Cabinet decisions on September 8, 2015, included: (1) financial measures: 1.1 low interest loans to provide operating cash flows for SMEs, 1.2 improved criteria and conditions of loan collaterals under PGS-5 project, and 1.3 support measures for SME through joint investment, (2) fiscal measures: 2.1 reduction of corporate income tax for owners of SMEs, and 2.2 tax measures to promote new start-ups.

Going forward, the Committee were of the view that monetary policy stance should continue to be accommodative to support the economic recovery. The Committee also placed emphasis on enhancing the effectiveness of monetary policy transmission, especially to the real economy, a task that requires continued monitoring and evaluation. In addition, the MPC will monitor imbalances that may build up as a result of a prolonged low interest environment.

Between the MPC meetings on June 10 and August 5, 2015, global and regional financial markets exhibited higher volatility. Contributing to increased volatility were mounting market expectations that the Fed would increase the policy rate by year-end, concerns over the Chinese economic slowdown and monetary policy direction, and uncertainties over the negotiation of Greece's bailout package. These factors led to rapid baht depreciation against the U.S. dollar. Meanwhile, Thailand's economic recovery confronted a growing number of unfavorable factors, particularly the Chinese slowdown, drought, sluggish demand, and persistently low oil prices. It is likely, therefore, to be later than previously thought before headline inflation turns positive.

Under these circumstances, the MPC voted unanimously on August 5, 2015, to leave the policy rate unchanged at 1.50 percent. The decision took into consideration the effectiveness and risks of additional policy easing amidst heighted global volatility. The MPC judged that overall monetary conditions remained accommodative. This was reflected by the exchange rate movements that stayed conducive to the economic recovery and firms' costs of funds in the money market that had declined following the previous policy rate reductions, despite

potentially limited and inert commercial banks' responsiveness to additional policy easing in the current situation, because banks remained cautious with regards to lending under current economic conditions. The Committee thus decided to maintain the policy rate to support financial market stability, while assessing evaluating the impacts from previous policy actions on the real economy.

Regarding long-term financial stability, some MPC members believed that maintaining the policy rate could prevent further build-up of imbalances in the economy. The low interest rate environment had persisted for a long period and signs of increased speculative activities in the real estate sector became more evident, albeit not at alarming levels. Moreover, sluggish economic recovery was due partly to structural problems, such as the lack of private sector confidence that weighed down investment, and declining competitiveness. Monetary policy may not be the right policy tools to address these structural constraints effectively.

Later at its September 16, 2015, meeting, the MPC voted unanimously to maintain the policy rate at 1.50 percent. The Committee still prioritized the need to preserve financial market stability, amidst greater global market volatility caused by developments in the Chinese financial markets, the renminbi devaluation, and uncertainty over the timing of the Fed's policy rate hike. In this light, the Committee placed emphasis on considering the different global financial market scenarios and possible consequences on domestic economy before making a policy decision. In the Committee's assessment, external stability was robust enough to absorb global market volatility. Strong external positions were reflected in the solid current

account surplus, ample official international reserves, moderate level of external debt, and sound management of exchange rate risk by businesses with foreign debt.

In terms of growth outlook, the MPC judged that the Thai economy will continue to be weighed down by negative factors, especially those associated with the global economy, and downside risks that were likely to subdue growth. Nevertheless, the overall monetary conditions, including the exchange rate, continued to remain conducive to the ongoing economic recovery. Some MPC members assessed that further reduction in the policy rate during this period was likely to have a marginal impact on the economic recovery, owing to financial institutions' caution to extend credits and structural shortcomings that could not be fixed by monetary policy. Moreover, further policy easing may lead to imbalances in the financial system over the long term.

Going forward, the Committee were in agreement that the Thai economy would continue to face downside risks and therefore warranted the need for adequate and ongoing support from monetary policy. In this regards, the Committee would stand ready to utilize an appropriate mix of available policy tools to manage the downside risks that could emerge in period ahead. The MPC also continues to evaluate and further enhances the effectiveness of monetary policy transmission in the current environment. In addition, monetary policy should play a role in shoring up market confidence, given that market sentiments have been quite sensitive to various developments, in order to ensure robust and stable recovery of the Thai economy.

1.3 Appendix: Tables for supporting assumptions and forecasts

Table 1.3 Forecasts for GDP and Components			
Percent (per annum)	2014*	2015	2016
GDP growth	0.9	2.7	3.7
Domestic demand	-0.1	2.9	3.4
Private consumption	0.6	1.9	2.7
Private investment	-2.0	-0.5	5.4
Government consumption	1.7	4.0	3.1
Public investment	-4.9	21.8	4.2
Exports of goods and services	0.0	0.6	3.3
Imports of goods and services	-5.4	0.4	4.4
Current Account (billion US dollars)	14.2	25.5	15.1
Value of merchandise exports	-0.3	-5.0	1.2
Value of merchandise imports	-8.5	-9.3	5.3

Note : *Outturn

Table 1.4 Forecast assumptions			
Annual percentage change	2015	2016	2017
Dubai oil price (U.S. dollars per barrel)	96.7	52.8	51.3
Non-fuel commodity prices (%YoY)	-3.9	-16.5	0.2
Fresh food prices (%YoY)	4.8	-4.0	3.5
Minimum wage in the Bangkok Metropolitan Region (baht per day)	300	300	300
Government consumption (current price) (%YoY)[1/]	3.4	6.9	5.9
Public investment (current price) (%YoY)[1/]	-4.7	19.2	6.9
Fed Funds rate (% at year-end)	0.13	0.38	1.38
Trading partners' economic growth (%YoY)[2/]	3.5	3.3	3.6
Regional currencies vis-à-vis the U.S. dollar (Index)[3/]	135.9	150.0	152.1

Note : [1/] Including spending on water management plans and infrastructure investment projects
[2/] Weighted by each trading partner's share in Thailand's total exports
[3/] Appreciation against the US dollar indicated by the minus sign

Table 1.5 GDP growth forecasts by research houses		
	2015	2016
HSBC	3.1	3.3
Maybank Kim Eng	3.0	4.5
TMB Bank	3.0	3.5
FPO[1/]	3.0	-
NESDB[2/]	2.7-3.2	-
Standard Chartered	2.9	4.0
DBS Bank	2.8	3.7
Kasikorn Research	2.8	3.2
Tisco Securities	2.7	4.0
BOT	2.7	3.7
Nomura	2.7	3.0
ANZ Bank	2.6	3.5
Kiatnakin Phatra Financial Group	2.5	3.5
JP Morgan	2.4	2.8
Siam Commercial Bank	2.2	3.0
Credit Suisse	2.0	3.0

Note: Compiled and published by Reuters on September 14, 2015, except:
[1/] Published on July 28, 2015
[2/] Published on August 17, 2015, with the release of GDP data for 2015 Q2
Presented in descending order of 2015's forecast

Table 1.6 Headline inflation forecasts by research houses		
	2015	2016
HSBC	-0.4	2.3
Kasikorn Research	-0.4	2.0
NESDB[1/]	(-0.7)- (-0.2)	-
DBS Bank	-0.6	1.8
ANZ Bank	-0.6	1.6
Standard Chartered	-0.6	1.4
FPO[2/]	-0.6	-
Kiatnakin Phatra Financial Group	-0.7	1.5
JP Morgan	-0.8	1.7
Tisco Securities	-0.8	1.7
Siam Commercial Bank	-0.8	1.3
Nomura	-0.8	1.2
BOT	-0.9	1.2

Note: Compiled and published by Reuters on September 14, 2015, except:
[1/] Published on August 17, 2015 with the release of GDP data for 2015 Q2
[2/] Published on July 28, 2015,
Presented in descending order of 2015's forecast

Table 1.7 Probability distribution of GDP growth forecast								
Percent	2015		2016				2017	
	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2
10-12	0	0	0	1	1	2	2	2
8-10	0	0	1	3	5	6	6	6
6-8	1	4	13	18	21	21	20	18
4-6	13	15	21	21	20	19	18	17
2-4	54	40	35	29	26	25	24	23
0-2	29	32	22	19	17	16	18	18
(-2)-0	2	9	7	7	7	7	9	10
< (-2)	0	1	1	2	2	2	4	5

Table 1.8 Probability distribution of headline inflation forecast								
Percent	2015		2016				2017	
	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2
6-7	0	0	0	0	1	1	1	2
5-6	0	0	0	1	2	3	4	4
4-5	0	0	1	4	6	7	8	9
3-4	0	0	4	9	12	13	14	15
2-3	0	1	11	16	19	19	19	19
1-2	1	7	21	21	21	21	20	19
0-1	9	20	26	21	18	17	16	15
(-1)-(0)	34	31	21	15	12	11	10	9
(-2)-(-1)	39	25	11	8	6	5	5	4
(-3)-(-2)	15	12	4	3	2	2	2	2
(-4)-(-3)	2	3	1	1	1	1	1	1

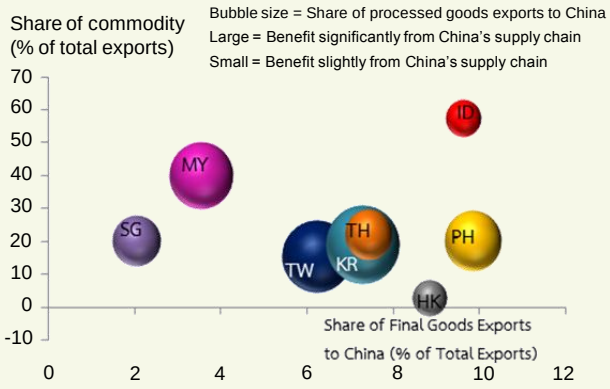
Table 1.9 Probability distribution of core inflation forecast								
Percent	2015		2016				2017	
	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2
3.0-3.5	0	0	0	0	0	1	1	2
2.5-3.0	0	0	0	1	1	2	4	5
2.0-2.5	0	1	2	4	5	7	9	11
1.5-2.0	6	9	8	12	13	15	16	17
1.0-1.5	34	27	21	21	21	21	21	21
0.5-1.0	44	35	29	25	24	22	20	19
0.0-0.5	14	21	24	20	19	17	15	13
(-1)-0.0	1	6	12	11	11	9	8	7
(-2)-(-1)	0	1	3	4	4	4	3	3
< -2	0	0	1	1	2	2	1	1

Effects of China's economic slowdown and the Renminbi devaluation on Thai exports

Economic data over the past year pointed to a slowdown in the Chinese economy. Flagging growth has been a result of the Chinese government's economic reforms, which aim at promoting domestic consumption as the main engine of growth in place of exports and investment, where both are projected to slow. The reforms also emphasize developing more sophisticated production technology along China's supply chain, thereby increasing China's capacity to produce import substitutes locally. Moreover, attempts to mitigate risks to financial and fiscal stability inevitably contributed to the slowdown during the transitional period. Financial markets generally perceived that the slowdown, to some degree, prompted the PBOC to devalue its CNY Fixing Rate on August 11, 2015, with an aim to bolster Chinese exports through weaker renminbi.

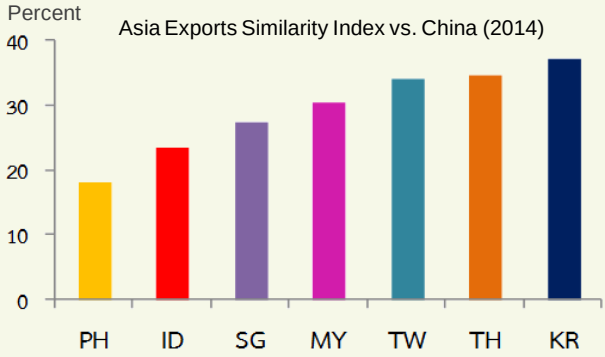
The renminbi devaluation raised concerns for the markets that the Chinese slowdown could be more severe than previously thought, and the Chinese currency could weaken further later this year. Several economies, closely connected with China, especially through trade links where China is both a partner and a competitor, will be affected. (1) **For China's trading partners, those with a high share of final products in their exports to China, and those with a high share of commodities in their exports,** particularly ASEAN countries, are likely to experience the most adverse effects. Their exports dipped because of the softening demand from China, higher import prices on the back of weaker renminbi, and declining world commodity prices. Nevertheless, those **countries with a high share of reprocessing products in their exports to China** will likely be cushioned from the impacts, on account of higher demands from China for their reprocessing products. This is due to stronger Chinese exports driven by increased price competitiveness following the renminbi devaluation **(Chart 1)**. **In addition, countries whose export products to China have low price elasticity (particularly those ASEAN countries** that export food, apparels and energy to China), will experience milder impacts than those whose exports have high price elasticity, such as smartphones and automobiles. (2) For **countries whose**

Chart 1 Impact on trading partners



Source: CEIC, Trademap, RIETI-TID 2013

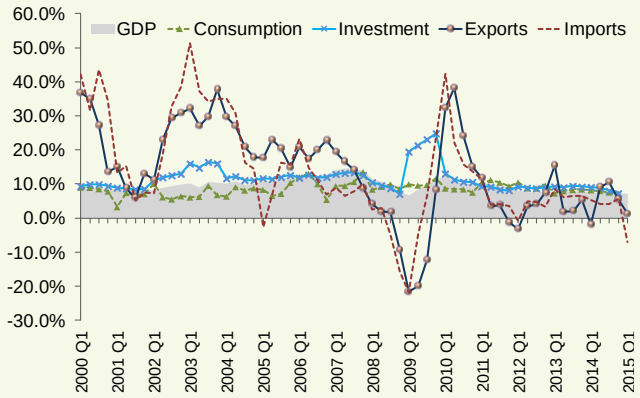
Chart 2 Impact on competitors



Source: Trademap, calculations by the Bank of Thailand

exports compete with China, those with high levels of the Export Similarity Index vis-à-vis China, particularly South Korea, Thailand, Taiwan and Malaysia, will experience higher impacts on their exports as a result of the renminbi devaluation **(Chart 2)**.

Chart 3 China's GDP growth, by component



Source: Oxford Economics

The effects of China's economic slowdown on Thailand were greater than previously assessed in the March 2015 Monetary Policy Report. Not only did Chinese growth fall by more than previously thought, but the causes of the slowdown also magnified the impacts on the Thai economy. That is, the slowdown was caused by weaker exports and investment which, compared to domestic consumption, were more closely linked to imports. In other words, **China's domestic-consumption-driven growth did not translate into greater imports to the same degree that the exports- and investment-led growth of the past did (Chart 3).** The effects on Thai exports come via two channels: 1) Effects on exports quantity, directly through exports to China and indirectly through Thai exports to countries that are highly dependent on commodity exports, and 2) Effects on export prices in terms of lower commodity and electronics prices in the world market. The details are as below.

1) Effects on exports quantity

1.1 Direct effect on Thai exports to China. **The slowdown led by an income effect in the form of weakened Chinese demand for goods used for domestic consumption (final products)**, which constitute an increasing share of Thai exports to China **(Chart 4)**. In 2014, final products accounted for 51.6 percent of the total value of Thai exports to China. However, most of these final goods were farm and processed farm products, such as rubber and rubber products, cassava, rice, fruits and sugar. Because China's domestic production of these goods remained insufficient to meet the local demand, the adverse effects on Thai exports of these final products was lessened. Meanwhile, **the price effect that the renminbi devaluation would have on Thai exports is expected to be limited** due to the fact that the baht had weakened by more than the renminbi since the beginning of the year (as of September 8, 2015, the baht had depreciated by 9.6 percent against the U.S. dollar, while the renminbi had fallen only by 2.6 percent against the U.S. dollar). Meanwhile, Thai exporters might face price renegotiations due to softer demands from China and weaker renminbi. Nevertheless, the baht depreciation should help boost exporters' liquidity, partly alleviating the impact of renegotiations.

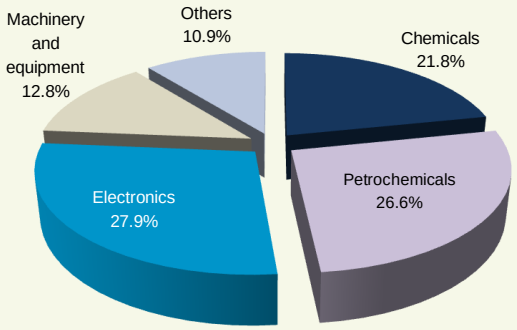
Chart 4 Share of Thai exports to China, by type



Source: Department of Custom, calculations by the Bank of Thailand

China's economic slowdown did not affect Thai exports of goods used in the manufacturing of Chinese exports (reprocessing products). These products, such as electronic, petrochemicals and chemicals, accounted for 48 percent of total exports to China **(Chart 5)**. Exports of these products were mainly dependent on demand from final destination countries, mostly G3 countries. However, **although Thai exports of these reprocessing products to China could gain from renminbi devaluation, the effect might not be as much as in the past.** This is a result of China's economic reforms which promote consumption of locally produced goods, thereby lowering Chinese demand for imports of this group of products from Thailand and other Asian countries in the China's supply chain.

Chart 5 Thai exports of raw materials and intermediate goods to China*, by type



Note: *Goods used by China for manufacturing of Chinese exports
Source: Customs Department, calculations by Bank of Thailand

Moreover, the effect of China's slowdown on Thai exports competing with Chinese exports to the third countries was likely to limited. Despite the fact that Thai and Chinese exports overlapped in many categories, including electronics, electrical appliances and machinery, closer analysis pointed to considerable differences between Thai and Chinese exports, both in terms of detailed product types and export markets. **(Chart 6)**. Therefore, **to Thailand China is considered more of a trading partner than a competitor.**

Chart 6 Structure of Thai and Chinese exports to third markets

	Products		Markets	
				
Electronics (14.6%)	Hard disk drive, integrated circuits and parts	Telecommunication equipment	The U.S. Hong Kong The E.U.	Hong Kong The U.S. South Korea
Electrical appliances (5.4%)	Air conditioners Refrigerators	Air conditioners Televisions	ASEAN The E.U. The Middle East	The U.S. Japan Mexico
Machinery and equipment (8.5%)	Compressors Auto parts Printers	Static converter Printers	ASEAN The E.U. Japan	Hong Kong The U.S. Japan
Processed farm products (11.9%)	Rubber products Canned fish	Paper Wood products	China The U.S. ASEAN	The U.S. Japan Hong Kong

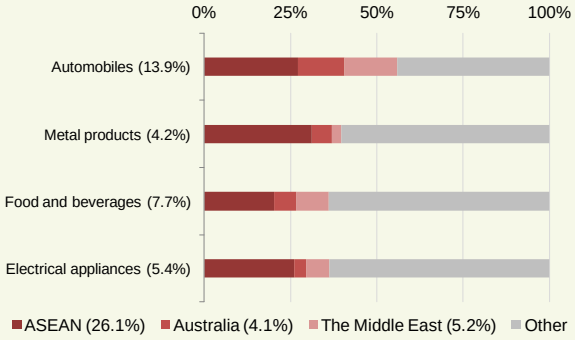
Note: (numbers in parentheses are share to total Thai exports in 2014)
Source: Trademap

1.2 Indirect effect through Thai exports to third countries, especially those that rely heavily on commodity exports. The slowdown in China, a major consumer of commodities, caused commodity prices to fall to the lowest levels in 6 years, since the subprime crisis of 2008. The decline in commodity prices hit countries whose exports comprise a high proportion of

commodities, such as the Middle East, Australia and ASEAN (especially Indonesia and Malaysia). This has led to weaker demand for Thai exports from these countries, especially automobiles, metal products, food and beverages, and electronic appliances **(Chart 7)**.

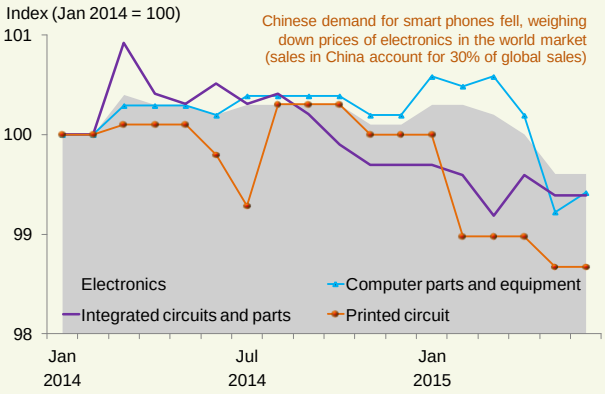
2) Effects on export prices. The fall in world commodity prices affected the prices of Thai exports, especially those that follow crude oil price, including petroleum, petrochemicals, chemical products, rubber and rubber products (these items accounted for 19.1 percent of total Thai exports in 2014). China's slowdown will likely cause commodity prices to remain subdued going forward. Moreover, China's recent expansion of production capacity in several industries, such as steel and plastic granules, could lead Chinese manufacturers to compete more on price in the world market in order to compensate for softer domestic demand. This could exert further pressure on prices in the global market.

Chart 7 Thai exports to countries that depend on commodities



Note: (numbers in parentheses are share to total Thai exports in 2014)
Source: Customs Department, calculations by Bank of Thailand

Chart 8 Price index of Thai electronics exports



Source: Ministry of Commerce

Moreover, China's slowdown led to falling global prices of electronics as China is a major consumer of electronic products (smart phone sales in China roughly totaled 30 percent of global sales, which contracted for the first time in 6 years in 2015 Q2). Electronics accounted for 14.6 percent of Thai exports, and their prices are set to decline steadily following downtrends in global prices. Interviews with Thai exporters of electronics confirmed the outlook, as Chinese buyers renegotiated for lower prices due to the weaker demand **(Chart 8)**.

In sum, the effects of China's economic slowdown and renminbi devaluation on the Thai and other Asian economies were more severe than earlier assessed. This is due to the larger-than-expected deceleration of China's growth, as well as the slowdown in Chinese investment and exports which, compared to consumption, were more closely linked to China's imports. For the Thai economy, the income effect of China's economic slowdown was greater than the price effect from renminbi devaluation. Quantity of exports also declined as a result of the drop in Thai exports to China and third countries, especially those with a high share of commodities in their exports. Export prices also decreased, particularly prices linked to crude oil price and electronics. **Taking into account of these factors, along with the shifts in global trade structure and persistent structural constraints for Thai exports, the Bank of Thailand has revised down its exports forecast in 2015. Specifically, Thai exports is projected to contract by 5 percent, worse than the earlier assessment of a 1.5-percent contraction provide in the June 2015 Monetary Policy Report. However, although the value of exports in U.S. dollar terms will most likely fall, exports measured in baht terms are expected to remain stable or edge up slightly due to the weakening of the baht against the U.S. dollar.**

Recent Economic Developments

2. Recent Economic Developments

The global economy is likely to expand at a slower pace than projected in the previous monetary policy report, following economic slowdown in China and other Asian economies. Advanced economies continued to post a steady growth, with the U.S. economy recording a solid expansion, the euro area economy showing pronounced signs of recovery, and Japanese growth likely to pick up gradually. However, downside risks to global growth heightened due to the slowdown of the Chinese and other Asian economies, which could be further impacted by global financial market volatility and renminbi devaluation.

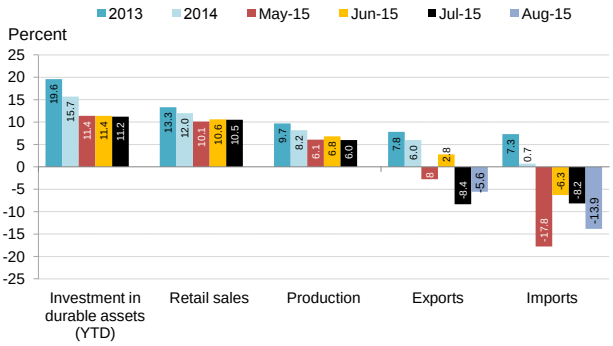
The Thai economy continued to recover at a gradual pace, mainly supported by tourism and public spending. Looking ahead, the Thai economy continued to face increased downside risks due to weaker growth in China and other Asian economies. Moreover, lower commodity prices weighed down Thai exports of goods and services, lowering employment, income and private sector confidence. As a result, households remained cautious with spending while firms delayed plans for investment.

2.1 The global economy

The global economy is likely to expand at a slower pace than assessed in the last monetary policy report due to projected slower-than expected growth in Asian economies.

The Chinese and other Asian economies (excluding Japan) expanded at a lower rate than previously assessed. In 2015 Q2, **the Chinese economy** grew by 7.0 percent (yoy), a similar rate with the previous quarter **(Chart 2.1)**. China's service sector registered a strong growth, especially in financial services. However, the manufacturing sector continued to slow down, owing mainly to sluggish investment in industries with excess production capacity and in the real estate sector burdened with a high level of excess

Chart 2.1 China's economic indicators
(percentage change from the same month last year)



Source: CEIC

supply. Increased stock market volatility since July 2015 **(Chart 2.2)**, together with weaker renminbi, could have adverse impacts on the real economy in the periods ahead, with the costs of production expected to rise due to higher import prices.

Nevertheless, in the first two months of Q3, the Chinese authorities introduced additional fiscal and monetary stimulus to boost growth, including (1) a policy rate reduction of 0.25 percent and an attenuation of the reserve requirement ratio (RRR) and (2) expedition of infrastructure investment and relaxation of local government's funding constraints. These measures were expected to facilitate a soft landing of the Chinese economy.

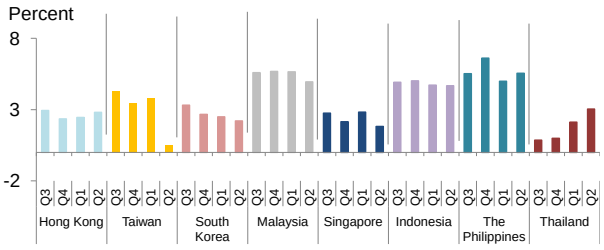
Growth in most **Asian economies** (excluding China and Japan) slowed in 2015 Q2 in line with contraction in exports and domestic demand **(Chart 2.3)**. Q3 data indicated that exports of Asian economies should continue to contract due to (1) weaker Chinese growth affected countries, which export a large share of final products to China and are net exporters of commodities, (2) economic activities within the region remained weak despite higher demands from the U.S., and (3) weaker renminbi could affect countries competing with China on similar export goods to the same destinations. Signs of sustained decline in exports of Asian economies (excluding China and Japan) had adverse impacts on domestic demand, particularly investment and manufacturing production. This was reflected by a steady drop in imports of machinery and equipment. At the same time, private consumption softened as households became more cautious with their spending amidst concerns over the prospects of weaker manufacturing employment.

Chart 2.2 China's stock market conditions
(Shanghai Stock Exchange)



Source: CEIC

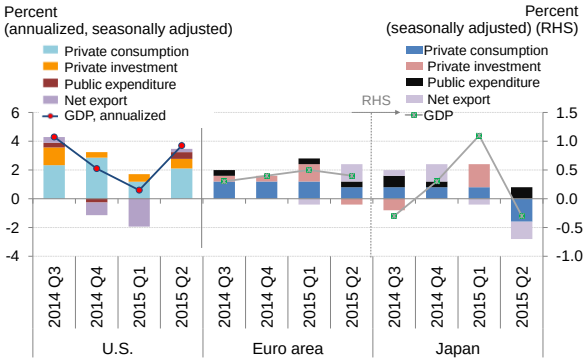
Chart 2.3 GDP growth of Asian economies
(change from the same quarter last year)



Source: CEIC

Advanced economies on the whole continued to expand, owing mainly to solid U.S. growth **(Chart 2.4)**. In 2015 Q2, **the U.S. economy** expanded by 3.7 percent (qoq saar, second estimate), up from 0.6 percent in the previous quarter. The economy was bolstered by stronger growth in private consumption, due to a steady recovery in the labor market, and by higher investment in machinery and equipment in line with rising business profits. Net exports registered positive growth, after having been disrupted by temporary factors, such as the dock workers protest at the West Coast ports. The U.S. economy should see continued expansion in Q3, as suggested by manufacturing production indices and retail sales.

Chart 2.4 Sources of GDP growth of G3 economies
(change from the same quarter last year)



Source: Bureau of Economic Analysis, Eurostat, Cabinet Office of Japan

The euro area economy[1/] grew by 0.4 percent (qoq sa) in 2015 Q2, slightly down from 0.5 percent in the previous quarter. Spain, Germany and Italy posted robust growth. Growth in the euro area economy was underpinned by stronger exports, which benefited from the weaker euro, and stronger private consumption on the back of low oil prices. Continued accommodative monetary conditions should help foster economic growth in Q3.

The Japanese economy contracted by 0.2 percent (qoq sa) in 2015 Q2, following a 1.1 percent growth in the first quarter, due mainly to a contraction in consumption caused by unusual weather, a temporary shock. Consumption is expected to edge up and provide support for growth in Q3. Preliminary economic indicators showed signs of improvements in Q3, such as real household spending index, retail sales and real

[1/] The euro area economy consists of 18 countries that share the euro as an official currency.

wages. Exports, however, remained sluggish in line with the slowdown in trading partners' economies.

Looking ahead, the Committee judged that the global economy would continue to recover, albeit at a slower pace due to the slowdown in Chinese and Asian economies. Downside risks to growth in trading partners' economies increased from the earlier assessment, reflecting higher risks linked to the Chinese economic slowdown and additional adverse impacts of weaker renminbi on exports of Asian economies. Moreover, developments on Greece's economic and fiscal reforms need to be monitored given the country's fragile economic fundamentals, although uncertainty over the financial rescue plan has somewhat subsided in the near term.

The Chinese economy is expected to expand at a slower pace due to softening investment, which is a result of China's rebalancing of the economic structure as well as financial and fiscal reforms. However, the government continued to stimulate the economy through investment in infrastructure and some primary industries. Consumption growth is likely to continue due to robust labor market, while exports are expected to gradually pick up in line with global recovery, especially in the U.S.

Asian economies (excluding China and Japan) are projected to moderate, due mainly to weaker exports. Export growth should pick up going forward due to demand from advanced economies. However, export growth is likely to be modest as growth in China and Asian economies are likely to remain soft throughout the forecast period.

The U.S. economy is expected to record a steady growth going forward, supported by improved fundamentals. Labor market is projected to recover in line with higher employment, while wages are likely to increase under tighter labor market conditions. Private sector's financial position should improve from continuing deleveraging. Notable risks in the future include (1) uncertainty over the Fed's interest rate tightening, leading to increased financial market volatility, (2) a slowdown in investment in the energy sector, due to persistently low oil prices, and (3) U.S. dollar appreciation hampering exports.

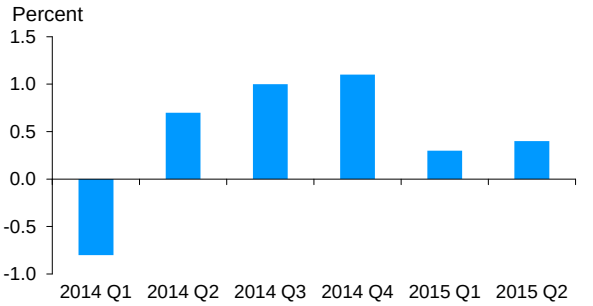
The euro area economy is projected to recover gradually, driven by accommodative monetary policy and low oil prices. Moreover, business and consumer confidence should be bolstered by the conclusion of negotiations between Greece and its creditors on debt repayment conditions and structural reforms required in August 2015 to secure the third financial rescue package. However, weakening global trade outlook is likely to weigh down exports.

The Japanese economy is expected to expand at a gradual pace. Stronger economic activities are supported by private consumption, due to higher household incomes following growing employment. However, export growth may be constrained by softer growth in China and other Asian economies.

2.2 The domestic economy

The Thai economy expanded slightly in 2015 Q2 from the last quarter **(Chart 2.5)**. Tourism and public spending were key drivers of growth. Over the same period, exports continued to contract due to the following two factors. (1) Trading partners' economies were adversely affected by the slowdown in the Chinese economy and the renminbi devaluation, and by plunging commodity prices that hurt export revenues of some trading partners and lowering prices of Thai exports. (2) Structural problems within the Thai exports sector and shifts in global trade structure both undermined the Thai export performance.

Chart 2.5 GDP growth[1/]
(seasonally adjusted, percentage change from last quarter)



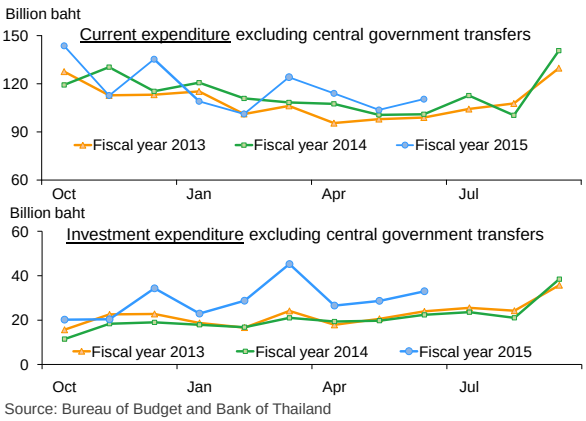
Note: [1/] Calculations based on chain volume measure (CVM)
Source: Office of the National Economic and Social Development Board

Downside risks from domestic factors are unlikely to have as much of an impact on the Thai economy as external risks. If violent incidents inside the country are contained, then the impact on tourism should only be short-lived. This year drought conditions are also beginning to dissipate. However, many potential risks, on the whole, continue to pose a threat to employment, income and private sector confidence, prompting households to restrict spending and firms to cut production and delay investment plans.

Public spending, especially investment expenditure, and tourism have been important drivers for the economy.

Public spending, both current and investment expenditures, continued to expand from the last quarter (Chart 2.6). Current expenditure (excluding government transfers) expanded, due to higher civil servant salaries and higher spending on supplies of educational agencies. Investment expenditure rose steadily,

Chart 2.6 Public spending



Source: Bureau of Budget and Bank of Thailand

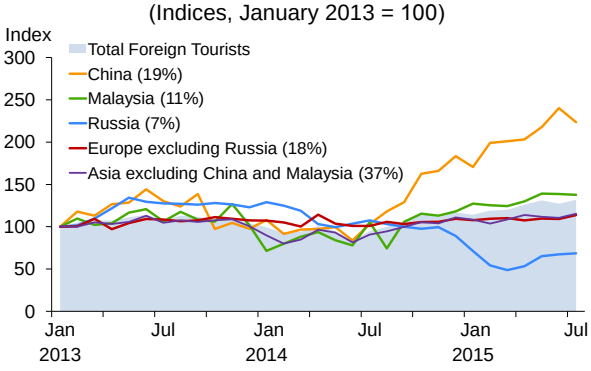
due to front-loading of investment projects. In July 2015, budget disbursements have begun for the phase II economic stimulus. Road transport infrastructures and water management system projects are likely to see rapid and continued disbursements because they consist mostly of small and medium projects with completion period under one year.

Going forward, public spending will continue to play a key supportive role for economic growth. The economic policy of the new cabinet focuses on boosting investment and employment, and bolstering private sector confidence. Nevertheless, infrastructure investment megaprojects which will help spur private investment, such as rail system development and inter-city motorway projects, are still in their initial phase and require some time to make progress.

Tourism continued to post a solid growth due to the growing number of Chinese and Malaysian tourists, and the steady rebound of Russian tourists arrivals **(Chart 2.7)**.

Looking ahead, tourism is projected to expand at a slower pace. Tourist arrivals at Thailand's major airports fell following the Ratchaprasong incident on August 17, 2015, but still recorded a higher growth than last year. Several countries had issued travel warnings urging their citizens to be more careful traveling in high-risk areas, but none issued a travel ban on Thailand. Based on past incidents, Asian tourists were most sensitive to unrests. However, when there were volatile situations in Bangkok, some tourists chose to change their travel plans to visit other parts of Thailand instead. Therefore, if the situation is well under control, past experience suggests that the tourism sector is likely be

Chart 2.7 Number of tourists by origins
(Indices, January 2013 = 100)



Note: Parentheses ( ) indicate shares of total foreign tourists in 2014
Source: Department of Tourism

affected only in the near term and should return to normalcy within 3 months **(Chart 2.8)**.

Service sector posted a solid growth following robust tourism

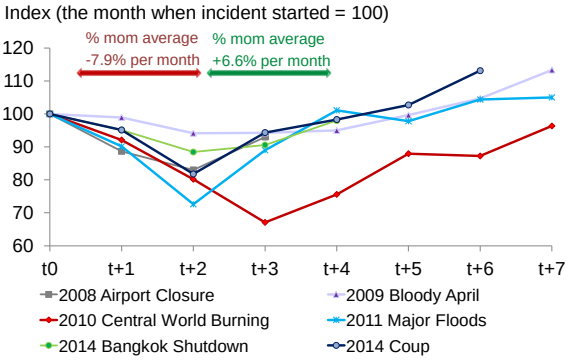
The service sector in part provided support for the economy. Hotels, restaurants, aviation transport, and non-durables businesses benefited from the rise in the number of foreign tourists. In addition, the telecommunications sector grew from increased demand, spurred by advancement in media and communications, together with service providers' promotional campaigns **(Chart 2.9)**.

Going forward, air transport service (which accounts for 2 percent of the service sector) faces increasing risks if the Thai aviation sector fails to meet the standards set by the International Civil Aviation Organization (ICAO). Revenue from air travels could slow over the period where the sector are unable to meet the required standards[2/]. Tourism revenue should not be much affected because the number of tourists is projected to continue rising, while travelers can choose other unaffected airlines.

Exports contracted in both quantity and price terms **(Chart 2.10)**

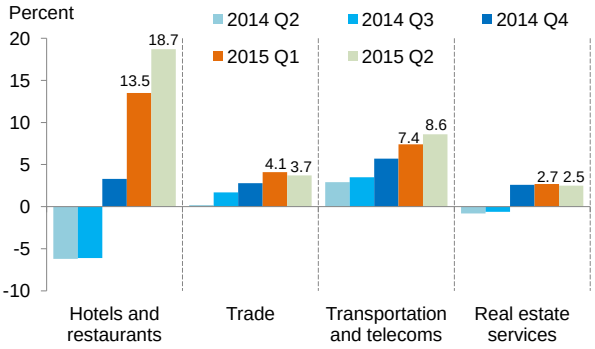
Exports contracted from the previous quarter due to weaker growth in trading partners' economies in Asia, the Middle East and Australia. Because commodities form a major part of exports of these economies, they suffered from declining

Chart 2.8 Index of foreign tourists following violent incidents



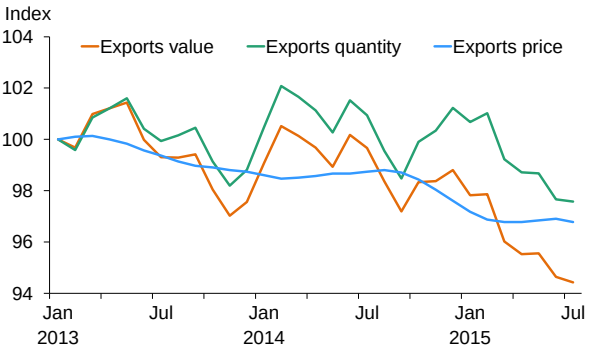
Source: Bank of Thailand

Chart 2.9 Growth of the service sectors



Source: Office of the National Economic and Social Development Board

Chart 2.10 Index of value, quantity and price of merchandise exports (3-month moving average, Jan 2013 = 100)

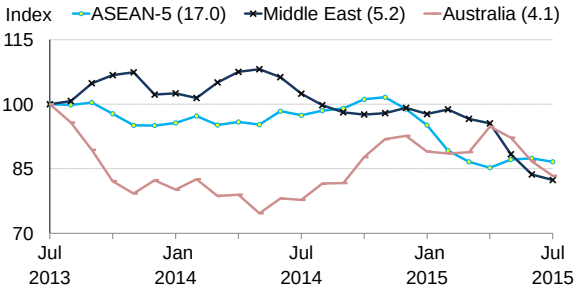


Source: Customs Department (data processed by Bank of Thailand)

[2/] Based on the experience of other countries, including the Philippines, Israel and India, it took 4-6 years for the ranking on credibility to be restored and for aviation standards to be improved.

commodity prices. Thai exports to these markets thus declined **(Chart 2.11)**, especially in automobiles, food and beverages, electrical appliances, and metal products. Exports of petroleum-related products also fell due to falling crude oil price. Moreover, production of some export items faced structural problems. Hard disk drives (HDD) manufacturing failed to adapt to the changing needs of today's consumers, while loss of competitiveness continually weighed down manufacturing and exports of labor-intensive goods, such as textiles and apparels. In addition, shifts in the global trade structure led to a decline in global trade, in part because some countries opted to produce intermediate goods locally to substitute for imports. However, the fall in automobile exports this quarter was caused by a temporary factor, as consumers chose to wait for new commercial car models, manufactured in Thailand.

Chart 2.11 Thai exports by destination markets
(seasonally adjusted, Jul 2013 = 100, 3-month moving average)



Note: ( ) indicates share of total Thai exports in 2014
Source: Customs Department (data processed by Bank of Thailand)

Looking ahead, **the Chinese slowdown and renminbi devaluation are expected to have more significant effects on Thai exports, both through direct and indirect impacts. The direct effects are likely to be smaller than indirect effects.** In term of direct effects, exports of final products for consumption in China have continually declined since last year. The decline has been mostly in farm produce including rubber, where there is lower demand amidst the economic slowdown and where China could partially produce locally. Indirect effects occur from the slowdown in trading partners' economies whose exports are adversely affected. Moreover, prices of Thai exports declined, especially farm and petroleum products, partly because of plunging global commodity prices *(see Box: The Effects of*

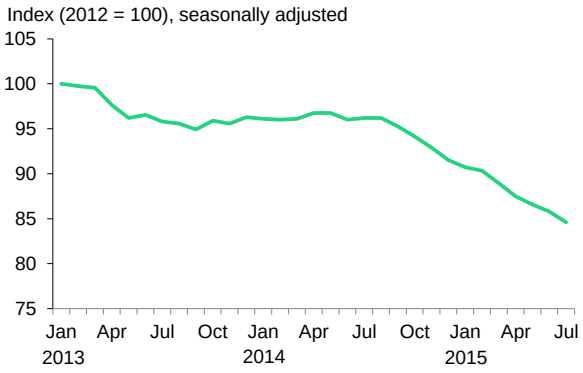
China's Economic Slowdown and Renminbi Devaluation on Thailand's International Trade).

Declining farm incomes and weaker consumer confidence prompted households to remain cautious with regards to spending

Private consumption edged up from the last quarter, mainly due to increased spending on non-durables necessities and on services, such as telecommunications-related expenses. However, spending on durables remained low because of last year's drought which reduced farm output and incomes **(Chart 2.12)**, together with depressed farm prices. Non-farm household income remained stable. Households postponed purchase of high-value goods due to (1) weaker private sector confidence **(Chart 2.13)**, (2) elevated household debt and (3) financial institutions' tighter lending standards for household credit following worsening loan quality.

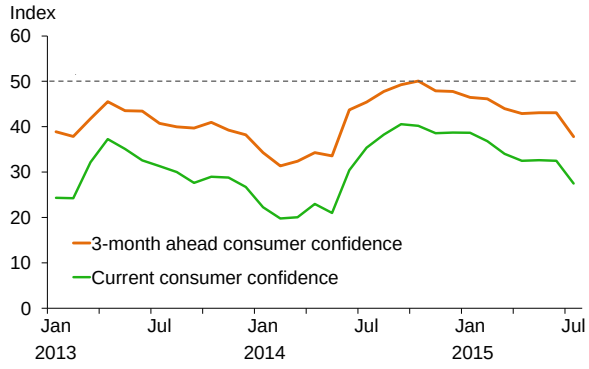
Going forward, private spending should recover slowly. Lower farm income and drought encouraged a migration of labor from farm sector to non-farm sector, but most workers moved into relatively low hourly wages jobs, such as construction, hotels and restaurants. Due to concerns over their future income, consumers remained careful with their spending. Furthermore, financial institutions have become slightly more cautious with regards to lending, due to concerns over economic outlook and the quality of consumer loans **(Chart 2.14)**. This year's drought has dissipated as rainfall picked up, but the late arrival of rainfall and the shorter-than-normal period of rain are likely to limit rice production in 2015 Q4. Moreover, water levels in important dams are still lower than past averages, which

Chart 2.12 Index of 12-month average farm income



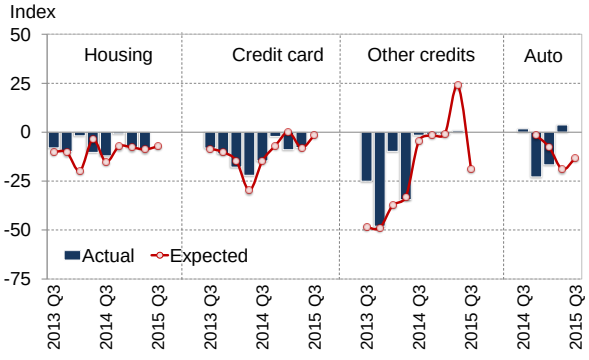
Source: Office of Agricultural Economics and Bank of Thailand

Chart 2.13 Consumer Confidence Index



Source: Ministry of Commerce, calculations by Bank of Thailand.

Chart 2.14 Lending standards to household sector



Note: Index > 0 indicates less strict standards; Index = 0 indicates stable standards; Index < 0 indicates stricter standards
Source: Survey of credit conditions, Bank of Thailand.

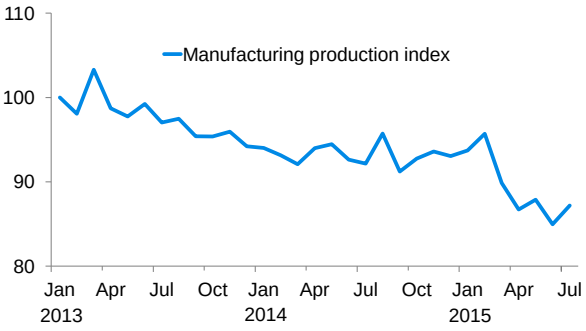
could undermine the recovery of farm incomes next year.

Weak domestic and external demand prompted firms to cut back production and delay investment plans for capacity expansion.

Manufacturing production fell from the previous quarter mainly from the decline in manufacturing exports (Chart 2.15). Output contraction occurred in the following industries **(Chart 2.16)**: (1) Commercial automobiles, as a major manufacturer was in the process of changing production line for new car models (a temporary factor). (2) Hard disk drive (HDD), as consumers switched to new hardware requiring computer components made of solid-state drive (SSD). (3) Food and beverages, due to a fall in beer production following a ramp-up in production in late March 2015, as well as a decline in frozen shrimp production resulting from a shortage of raw shrimps and continued decline in shrimp prices (due to offloading of shrimp stocks from India and Indonesia). (4) Textiles and apparels, due to high wages and constraints on production technology, as well as the European Union's withdrawal of GSP privileges which undermined price competitiveness of Thai products relative to those made by competitors.

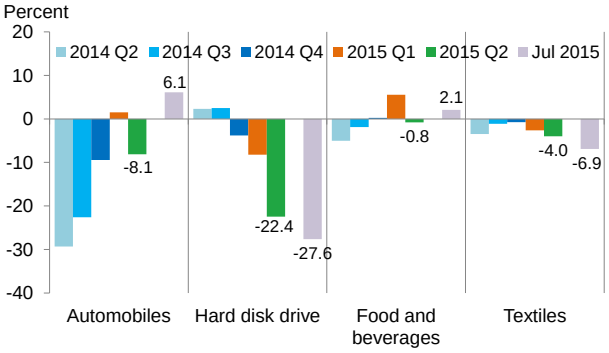
Chart 2.15 Manufacturing production conditions

Index (Seasonally adjusted, January 2013 = 100)



Source: Office of Industrial Economics, Ministry of Industry, National Statistical Office, The Customs Department, The Thai Automotive Industry Association, and the Bank of Thailand

Chart 2.16 Manufacturing production by sectors (compared with the same period last year)



Source: Office of Industrial Economics, Ministry of Industry

Going forward, automobile production is expected to grow from new car models. However, overall manufacturing production is likely to remain sluggish, particularly manufacturing for exports, due to feeble global demand.

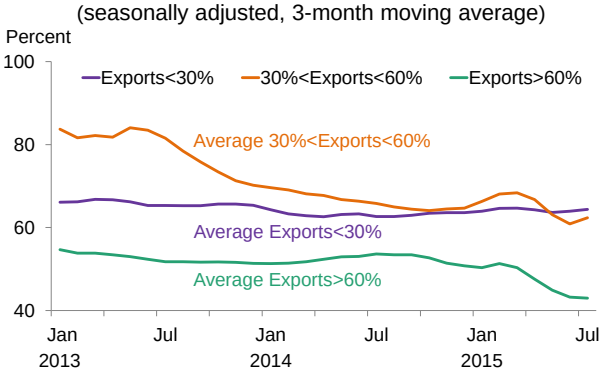
Private investment fell from the last quarter from weak domestic and external demands. Most businesses invested in domestic machinery mainly to repair existing machines and improve production efficiency. Imports of capital

goods and indicators on construction investment weakened from the previous readings. Investment for production capacity expansion remained sluggish, as the business sector still had ample spare production capacity, particularly manufacturers for exporting goods (exports accounting for over 60 percent of total output) whose capacity utilization was lower than other sectors **(Chart 2.17)**. The private sector also awaited greater clarity on economic recovery and government plans on large-scale infrastructure investment. This is consistent with declining private sector confidence following a drop in sales orders **(Chart 2.18)**.

Imports dropped in line with subdued production and investment

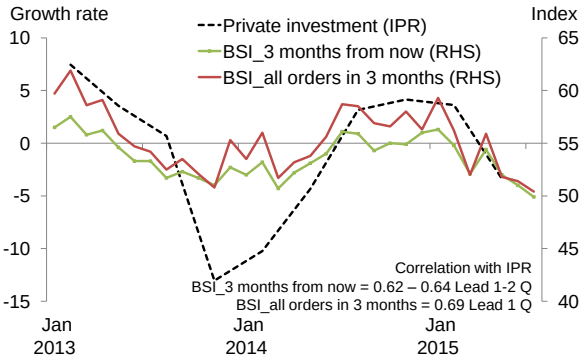
Imports fell from the previous quarter (Chart 2.19). Imports of raw materials and intermediate goods, excluding fuels, shrank. Contributing to lower imports were a fall in machinery imports consistent with subdued manufacturing production, as well as existing spare production capacity for the business sector. Imports of fuels picked up somewhat but remained at a low level in line with crude price. Imports of consumption goods moderated as domestic demand has yet to fully recover.

Chart 2.17 Production capacity by exports
(seasonally adjusted, 3-month moving average)



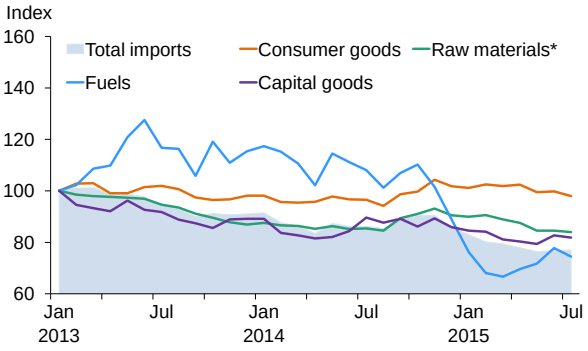
Source:Office of Industrial Economics, Ministry of Industry

Chart 2.18 Private investment outlook



Source: Office of the National Economic and Social Development Board, Business Sentiments Survey by Bank of Thailand

Chart 2.19 Imports index, by product groups
(seasonally adjusted, 3-month moving average, Jan 2013 = 100)



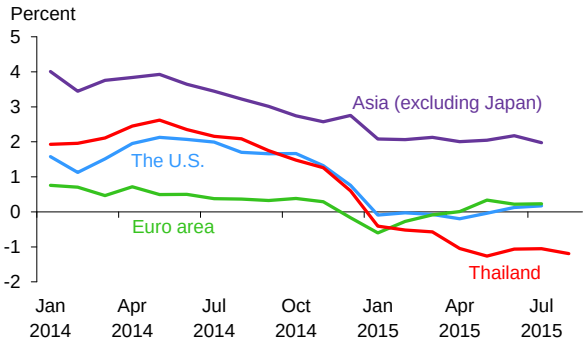
Note: *Exclude fuel and auto parts
Source: Customs Department (data processed by Bank of Thailand)

2.3 Production cost and price conditions

Inflationary pressures were generally weak, owing to declining domestic energy prices consistent with lower crude oil price, and softer domestic demand. Looking ahead, the MPC expected headline inflation to remain negative throughout the year, before rising gradually and turning positive in 2016 Q1, as the base effects of higher oil price in 2014 disappear. Core inflation is projected to remain positive at low levels throughout the forecast period. Deflation risk is not at a worrying level.

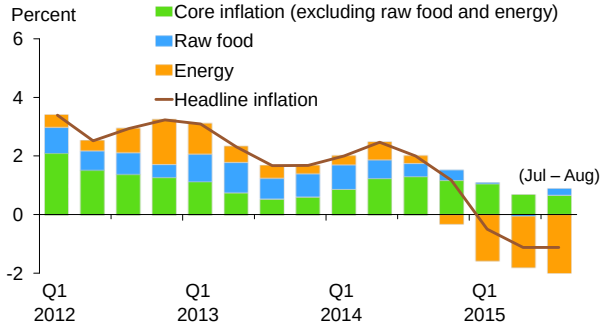
Global crude oil price fell recently, due to supply glut and lower demand. Global oil supply rose significantly because of increased production by OPEC member countries and the U.S., as well as the lifting of sanctions on Iran. Demand for oil softened as a consequence of China's economic slowdown. The resulting decline in oil prices dampened inflation readings around the world **(Chart 2.20)**. Negative headline inflation persisted in Thailand, averaging at -0.89 percent in the first 8 months of 2015, owing mainly to subdued crude oil prices **(Chart 2.21 and Table 2.1)**. Baht depreciation helped to soften the fall in the general price level through higher cost of imported raw materials especially crude oil, but could not fully prevent the decrease in the price level.[3/]

Chart 2.20 Domestic and foreign headline inflation



Notes: Foreign headline inflation calculated from unweighted averages
Source: Bureau of Trade and Economic Indices, Ministry of Commerce, CEIC, and calculations by Bank of Thailand

Chart 2.21 Contributions to headline inflation



Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by Bank of Thailand

[3/] A study on the impacts of exchange rate on inflation finds that baht depreciation affects headline inflation mainly through imports of crude oil, and only has a marginal effect on core inflation through imports of other raw materials. A 1-percent fall in the NEER causes headline and core inflation to rise by 0.07 and 0.03 percent within a year respectively.

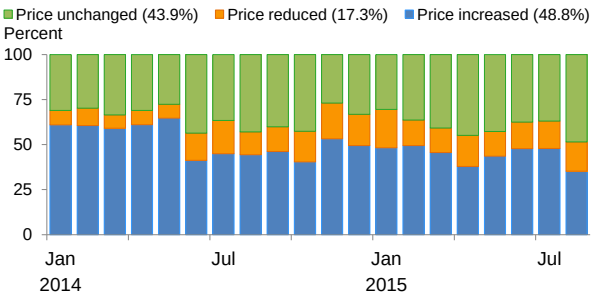
Table 2.1 Quarterly inflation

Unit: percent	2013	2014	2015				
			Q1	Q2	Jul	Aug	Jan-Aug
Percentage change from previous year (%Δyoy)							
- Headline Consumer Price Index	2.18	1.89	-0.50	-1.12	-1.05	-1.19	-0.89
• Core Consumer Price Index	1.00	1.59	1.47	0.96	0.94	0.89	1.14
• Raw food	5.54	3.46	0.30	-0.40	1.27	1.64	0.32
• Energy	4.79	1.68	-12.95	-14.16	-15.65	17.04	-14.25
Percentage change from previous quarter (%Δqoq_sa)							
- Headline Consumer Price Index	-	-	-0.8	-0.1	-	-	-
• Core Consumer Price Index	-	-	0.3	0.1	-	-	-
• Raw food	-	-	0.1	-0.5	-	-	-
• Energy	-	-	-8.9	-0.8	-	-	-

Source: Bureau of Trade and Economic Indices, Ministry of Commerce. Calculations of percentage change from last quarter, seasonally adjusted, by Bank of Thailand

The fall in the global crude price lowered domestic energy costs, including prices of retail oil, LPG for transport and households, and electricity. Since the beginning of the year, energy prices have on average dropped by 14.25 percent compared to the same period last year, thereby reducing some transportation and production costs. This, in turn, led to lower prices of other goods and services, especially the costs of public transport, where energy costs constitute 36 percent of total costs. Nevertheless, there have been no clear downward adjustments of prices of goods and services other than public transport, because energy costs only account for 7 percent of total production costs.[4/] This is consistent with changes in the prices of goods and services in the inflation basket, which reflected unchanged prices of most goods and services **(Chart 2.22)**. Based on surveys with business operators, energy and

Chart 2.22 Price adjustments and their shares in inflation basket



Note: Price changes calculated from %MoM data () is average share during 2010-2014
Source: Ministry of Commerce, and calculations by Bank of Thailand

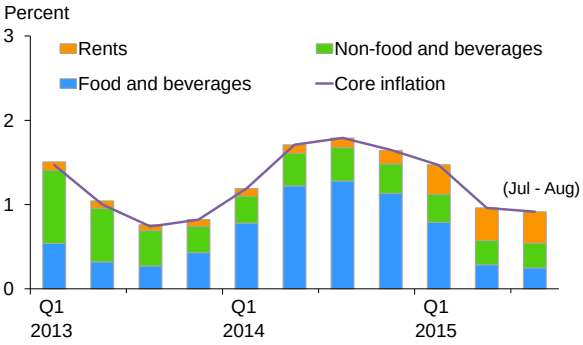
[4/] Cost ratio calculated from 2005 input-output tables.

transport costs were a minor factor influencing pricing decisions, relative to the cost of raw materials, wages and management costs. Business operators also tried to maintain their overall revenue amidst subdued domestic demand. However, some business operators adjusted prices downwards and competed more aggressively on prices to boost sales.[5/]

Aside from lower cost pressure resulting from lower crude oil price, demand-side pressure remained weak. Purchasing power was reduced by high household debt and lower incomes from both farm and non-farm sectors, owing to weak exports. Exports were held back by the slowdown in China and other Asian economies, depressed rubber prices in line with crude price and softer demand from China, as well as drought leading to lower rice output.

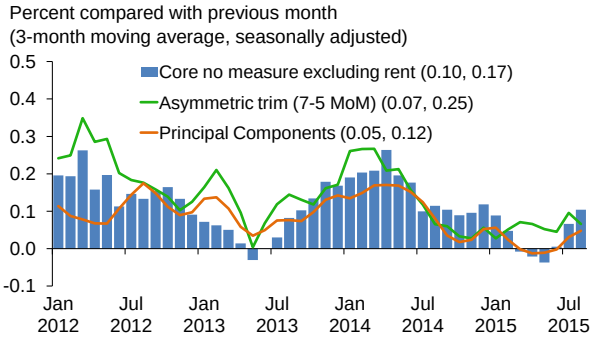
Low costs and weaker domestic demand caused core inflation to remain at low levels, averaging around 1.14 percent for the first 8 months of 2015. Food prices declined from lower raw food costs and the effect of a high base in the same period last year. Non-food prices fell in line with energy costs and softer domestic demand **(Chart 2.23)**, consistent with all three inflation indicators which were below their past averages[6/] **(Chart 2.24)**. Short-term inflation expectations edged down from significant decline in global crude oil price **(Chart 2.25).** However, deflation risk level is not deemed concerning as negative headline inflation was primarily caused by lower oil

Chart 2.23 Contributions to core inflation



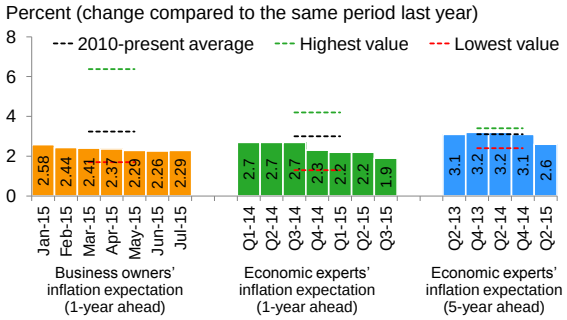
Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by Bank of Thailand

Chart 2.24 Underlying Inflation



Notes: Inside parentheses, figures on the left indicate Aug 2015 %MoM_SA (3mma), while those on the right show long-term average (2005-2014).
(Read more in FAQ Issue 93, Underlying Inflation, on the BOT Website)

Chart 2.25 Public inflation expectations



Note: Business owners' inflation expectation surveyed by Bank of Thailand
Economic experts' inflation expectation surveyed by Consensus Economics
Source: Business Sentiment Index Survey by Bank of Thailand and Asia Pacific Consensus Forecast

[5/] Behavioral survey on how business operators set wages and prices & Economic and business information exchange project between BOT and private firms.

[6/] Inflation indicators edged up slightly in the latest period, due to an increase in private school fees following the decision of the Private Education Commission.

prices, while public inflation expectations remained close to the inflation target, despite edging down slightly.

The MPC assessed that headline inflation should remain negative throughout the year and pick up slowly. Technically, headline inflation is not officially considered to be missing the target because the target is the 12-month average. However, based on existing forecasts on the economy and oil prices for the remainder of the year, the 12-month average of headline inflation in 2015 is expected to fall below the lower band of the target. Nevertheless, headline inflation is projected to turn positive in 2016 Q1 due to the diminishing base effect of high oil price. Meanwhile, core inflation is forecast to be stable throughout the projection. The MPC would continue to closely monitor inflation developments and continuously communicate with the public on inflation outlook.

Monetary and Financial Stability

3. Monetary and Financial Stability

Overall monetary conditions continued to be accommodative. Private sector's funding costs via commercial banks stabilized after declining in 2015 Q2. However, funding costs in the capital market fell in tandem with government bond yields, despite more recent increases following sovereign bond sell-offs in emerging markets. The Thai baht continued to weaken, as a result of expectation on federal funds rate hike, PBOC's recent renminbi fixings, and forecasts of slower-than-expected economic recovery for Thailand.

The fragile economy worsened financial positions of households and businesses, especially small firms, thereby lowering income and debt repayment ability of private sector. Loan quality of commercial banks deteriorated as a result. Nonetheless, commercial banks remained robust, owing to high level of loan loss provision and capital base. For the real estate sector, rising excess supply of condominiums requires close monitoring. Fiscal stability remained robust, with sufficient level of treasury cash balance and manageable level of public debt. Although public debt is set to rise due to new borrowings to cover budget deficit, the funds will be invested to enhance the economy's long-term potential.

3.1 Financial markets

Global financial markets

Volatility of the global financial markets rose substantially as a result of uncertainties over (1) the timing of Federal Funds' rate hike since the beginning of the year, (2) the resolution to the Greek soverign debt crisis between late March and July, (3) China's economic conditions and its stock markets, and (4) the devaluation of renminbi in August. The surge in volatility was reflected in the stock market, currency market, and bond

market, as indicated by the VIX (Volatility Index)[1/], CVIX (Currency Volatilty Index)[2/], and MOVE (Merrill Lynch Option Volatility Estimate)[3/] **(Chart 3.1).**

Thailand's financial markets

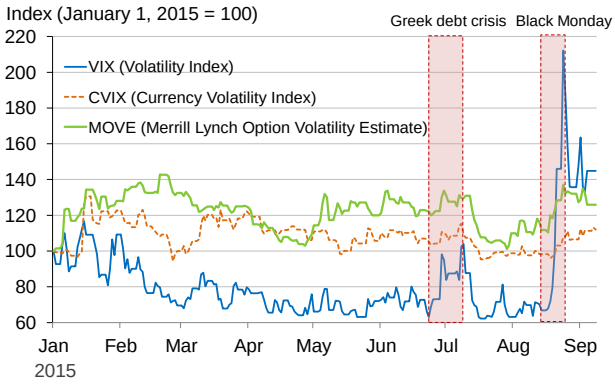
Funding cost of business in the capital market fell in tandem with lower government bond yields, despite recent increases following sell-offs of sovereign bonds in emerging markets. Meanwhile, money market rates stabilized, as policy interest rate was left unchanged.

Money and bond markets

From June to mid September 2015, money market rates stabilized, as the Monetary Policy Committee maintained the policy interest rate at 1.50 percent per annuum on June 10 and August 5 **(Chart 3.2).** The yields on short-term government bonds (up to 2-year sector) declined initially due to market expectation that the policy rate would be lowered further, given the weak economic recovery **(Chart 3.3).** Nevertheless, the yields drifted slightly upward from mid- August 2015.

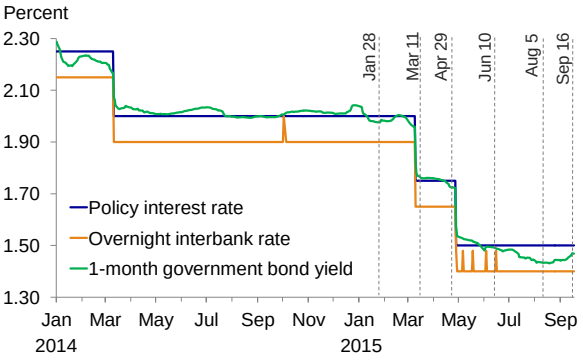
The yields on government bonds of medium to long maturities fell steadily from June to mid-August 2015. Factors that contributed to the decline include (1) the fall in yields of U.S. treasury securities (2) portfolio reallocation from risky assets to government bonds during the risk-off period triggered by the PBOC's adjustments of

Chart 3.1 Volatility indices



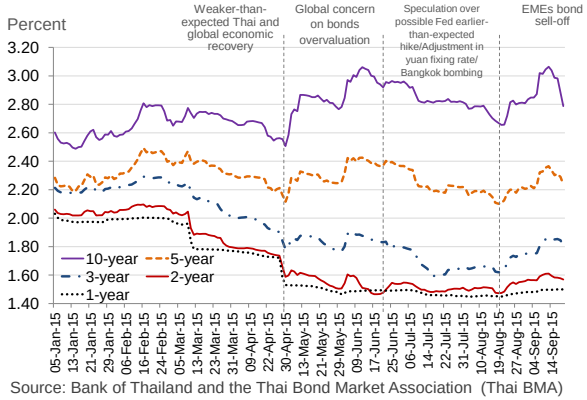
Source: Bloomberg

Chart 3.2 Money market interest rates



Source: Bank of Thailand and the Thai Bond Market Association (Thai BMA)

Chart 3.3 Government bond yields in 1 to 10-year sector



Source: Bank of Thailand and the Thai Bond Market Association (Thai BMA)

1/ VIX is an indicator for stock market volatility, measured by implied volatility of option prices of the S&P 500.

2/ CVIX is an indicator for currency volatility, measured by implied volatility of 9 major currencies.

3/ MOVE is an indicator for bond market volatility, measured by implied volatility of U.S. treasury bonds.

renminbi fixings, and (3) the incident at the Ratchaprasong intersection in Bangkok. Since mid-August, however, as investors' concerns over Chinese economic recovery intensified, investment was pulled out of emerging markets' bonds, and back into U.S. treasuries (flight to quality), causing Thai bond yields to rise in line with regional markets. Overall, compared to the June 10, 2015 MPC meeting, funding costs in the bond market declined **(Chart 3.4 and 3.5)** with the drop in bond yields, while credit spreads remained largely stable.

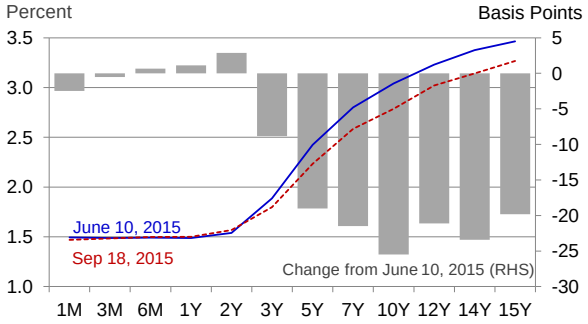
Going forward, the costs of funding in the bond market will likely fluctuate with external factors, particularly the initial increase in the Federal Funds'rate, as well as the uncertainty and divergence over monetary policy directions among major economies. These are key issues that the MPC continues to closely monitor.

Equity market

Stock market became less overheated as a result of net sales by non-resident investors in emerging markets, including Thailand. However, the decline in the stock market is expected to pose marginal risks to financial stability, because only a small fraction of trading was funded by margin loans.

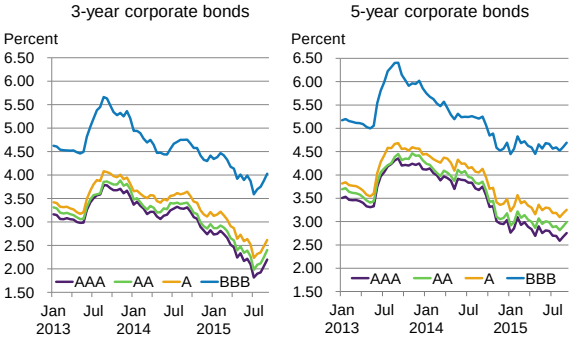
Equity market overheating continued to modertate since 2015 Q2 as a result of two main factors. The first factor was net sales by non-resident investors in emerging markets, including Thailand, in response to worsening economic outlooks for emerging economies especially China, and the expectation that the Federal Funds' rate will be raised by the end of this year. The second factor concerns domestic conditions that may delay Thailand's economic recovery.

Chart 3.4 Changes in government bond yield curves



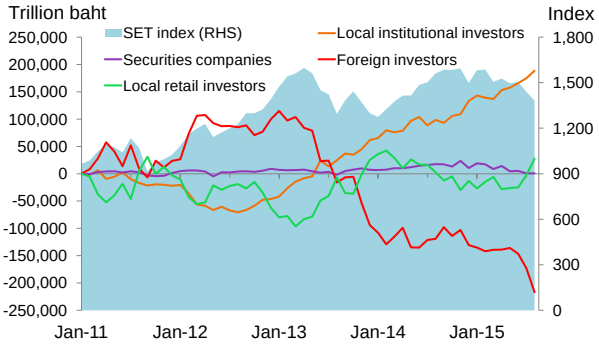
Source: Bank of Thailand and the Thai Bond Market Association (Thai BMA)

Chart 3.5 Thai corporate bond yields



Source: Thai Bond Market Association (Thai BMA)

Chart 3.6 The Stock Exchange of Thailand index and net buy classified by investor types



Source: The Stock Exchange of Thailand (latest dataon September 7, 2015)

Between June and August 2015, the SET and the MAI indices fell and foreign investors were net sellers in the SET **(Chart 3.6)**, driven by the above mentioned concerns. At the same time, investment by retail investors in the MAI contracted **(Chart 3.7)**.

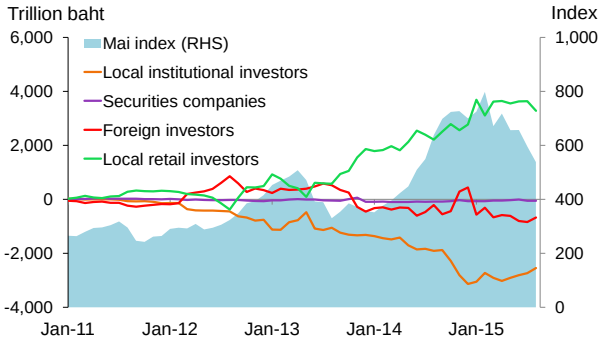
Despite the recent falls, stock valuations are considered to be elevated. The P/E (Price-Earnings Ratio) ratios in both markets remained above past historical averages and higher than those in a number of markets in the region **(Chart 3.8).** However, risks to financial stability are limited, due to low levels of trading funded by margin loans.

Foreign exchange market

The baht and most regional currencies weakened persistently against major currencies since mid-July 2015 **(Chart 3.9).** The weakness can be attributed to the market expectation on a growing likelihood that the Fed will start to raise its interest rate by the end of this year. Moreover, developments of the Greek debt negotiation, delay in domestic economic recovery, and concerns over severe draught added to the depreciation pressure.

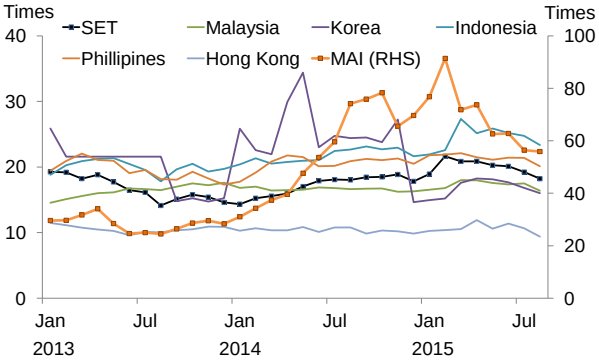
From August to early September 2015, the baht depreciated steadily on the back of three imporant events: (1) the adjustment of renminbi Fixing Rate by the PBOC since August 11, 2015, sending the baht and regional currencies weaker, (2) incident in Bangkok, (3) sell-offs of stocks and bonds in Thailand and other countries in the region by foreign investors, triggered by growing concerns over the recovery of Chinese economy and conflicts in the Korean peninsula. However, the decision by Federal Open Market Committee (FOMC) to maintain the fed funds' rate at the

Chart 3.7 The Market for Alternative Investment (MAI) and net buy classified by investor types



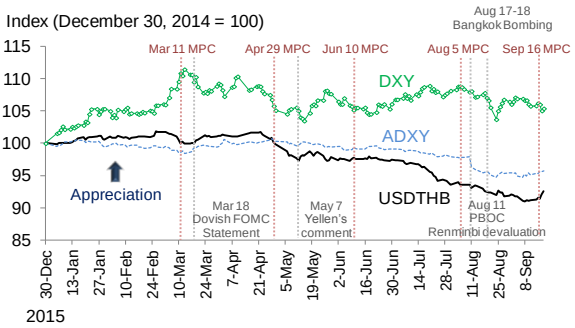
Source: The Stock Exchange of Thailand (latest data on September 7, 2015)

Chart 3.8 P/E ratios of regional stock indices



Source: The Stock Exchange of Thailand (latest data on September 7, 2015)

Chart 3.9 Movements in USDTHB and exchange rate indices

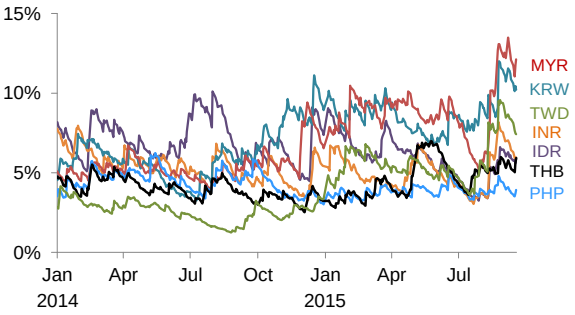


Note: DXY is an index of USD compared to a basket of EUR JPY GBP CAD SEK and CHF
ADXY is an index of Asian currencies (CNY KRW SGD HKD INR TWD THB MYR IDR and PHP) against USD
Source: USDTHB index from the Bank of Thailand, DXY and ADXY from Bloomberg

September 16-17 meeting moderately propped up the value of baht. On September 18, 2015, the baht per U.S. dollar stood at 35.53, down by 5 percent from the end of the second quarter. Further, the volatility of the baht rose slightly in this period, but remained low compared to those of regional currencies **(Chart 3.10).**

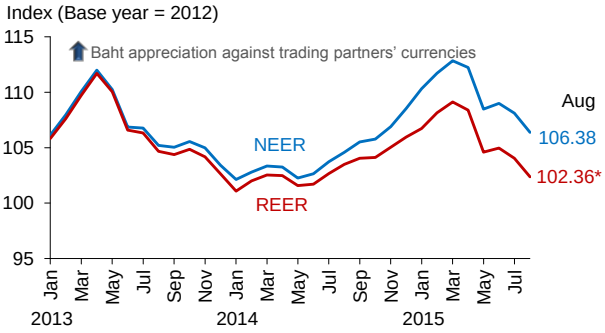
The depreciation of the baht against major currencies led to a 3.2 percent decline in the Nominal Effective Exchange Rate (NEER) in August 2015, compared to the average for the previous quarter. Meanwhile, the Real Effective Exchange Rate (REER) moved in line with the NEER **(Chart 3.11).**

Chart 3.10 Volatility of the baht and regional currencies



Note: Volatility is computed by Exponentially Weighted Moving Average (EWMA) method
Source: The Bank of Thailand and Reuters

Chart 3.11 NEER and REER



Note: *Preliminary data using NEER from August 2015, and inflation from July 2015 as a proxy for August 2015 inflation
Source: Bank of Thailand

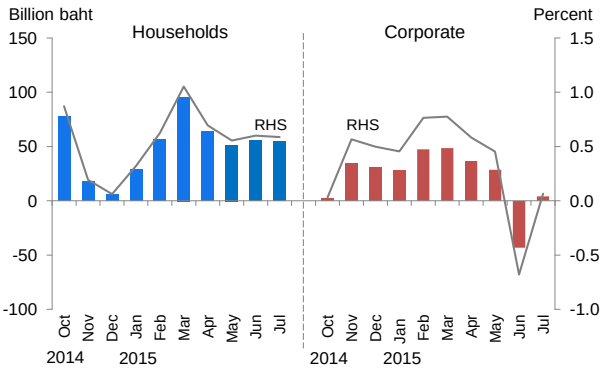
3.2 Financial institutions

Private sector funding costs and returns on deposits declined in line with the MPC decisions to maintain policy rate. Demand for funding also dipped during this period. Also, sluggish demand for loans led to a reduction in competition for deposits and new funds among financial institutions.

Interest rates, credits and deposits

With policy interest rate unchanged from 2015 Q2 until mid-September for the June 10, August 5, and September 16 MPC meetings, funding costs of commercial bank loans stabilized **(Table 3.1)**. Actual funding costs, however, depended on each loan applicant's credit risk, as commercial banks tightened their lending standards for both business and household loans. Between May and July 2015, household credit growth remained largely stable from 2015 Q1

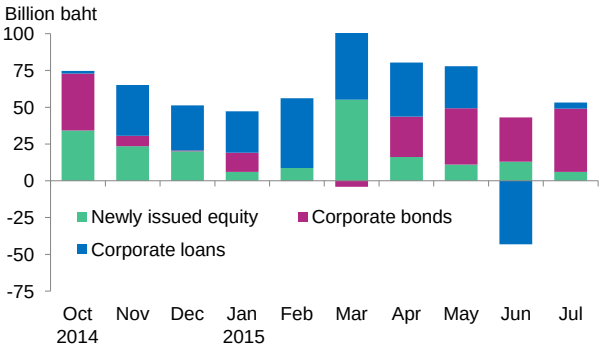
Chart 3.12 New private credits
(Changes in credit balance, seasonally adjusted)



Source: Bank of Thailand

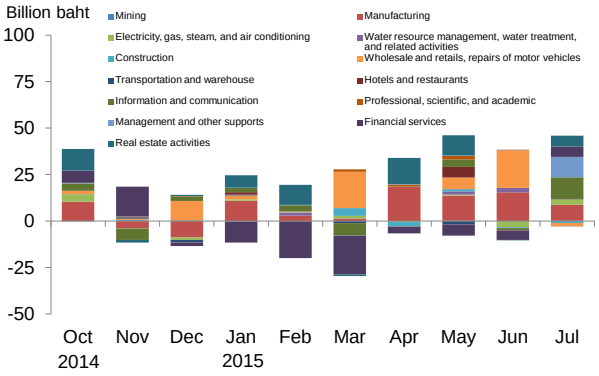
(Chart 3.12). Financial institutions were cautious in lending, because of elevated household debt and subdued farm incomes. Despite some signs of recovery during 2015 Q1, new issuance of credits to the private sector declined during this period **(Chart 3.12)** due to the following three factors. (1) A growing proportion of large firms, particularly in telecommunication and manufacturing sectors, increasingly relied on issuance of corporate bonds **(Chart 3.13)**, while in real estate sector the issuance had started to ebb **(Chart 3.14)**. (2) Credits to medium and small firms contracted, given financial institutions' tight lending standards and declining demand for credits. (3) A number of large firms repaid their loans. Overall, total private sector funding declined from 2015 Q1, as new bond issuance fail to offset the decrease in private credits.

Chart 3.13 Corporate financing*



Note: *Newly issued equities, change in corporate loans (seasonally adjusted) and corporate bonds
Source: Bank of Thailand and Thai BMA

Chart 3.14 Corporate financing via bonds



Source: Bank of Thailand and Thai BMA

Chart 3.1 Benchmark loans and deposits rates of commercial banks*

Percent	2013	2014				2015		
		Q1	Q 2	Q3	Q4	Q1	Q2	Sep 18
12-month deposits								
Average of the 4 largest commercial banks**	2.23	1.74	1.73	1.73	1.73	1.53	1.50	1.50
Average of remaining banks	2.61	2.35	2.22	2.19	2.22	1.93	1.75	1.72
Minimum lending rate (MLR)								
Average of the 4 largest commercial banks	6.84	6.75	6.75	6.75	6.75	6.63	6.51	6.51
Average of remaining banks	7.62	7.53	7.51	7.45	7.44	7.37	7.32	7.31
Minimum retail rate (MRR)								
Average of the 4 largest commercial banks	7.96	8.02	8.05	8.08	8.08	8.08	7.86	7.87
Average of remaining banks	8.77	8.70	8.70	8.63	8.63	8.62	8.54	8.54

Note: * Benchmark rates averaged across commercial banks at the end of the period
** Four largest commercial banks namely BBL, KTB, KBANK and SCB
*** Since August 2014, Bank of China (Thailand) became a domestically-registered commercial bank. Therefore, its interest rates are included in the Average of remaining banks rates in the table above.

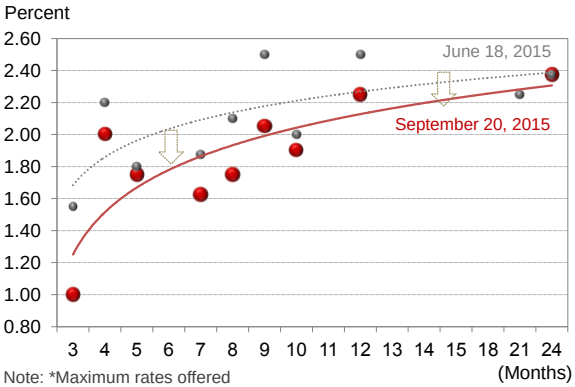
Given the backdrop of slowing momentum of private credits, commercial banks reduced special deposit products and steadily lowered deposit rates on these products **(Chart 3.15)**. Instead, some banks turned to mutual funds and insurance products in order to maintain their market shares. As a result, household and business deposit growth was lower from 2015 Q1 **(Chart 3.16)**, prompting a rise in loan to deposit (including bills of exchange (B/E)) **(Chart 3.17)**. Nonetheless, reduced liquidity of commercial banks did not impact their lending capacity, because the reduction in deposit was part of the banks' own adjustments to stalling credit growth, brought about by weak economic outlook.

The Senior Loan Officer Survey for 2015 Q2 indicated that credit demand from businesses will stabilize in the near future, while demand from households will fall, especially for home loans. Meanwhile, financial institutions will continue to tighten credit standards, especially for small and medium businesses and households. In order to assess the effectiveness of monetary easing, the MPC continues to carefully monitor the momentum of private sector funding. Such understanding will constitute one of the factors aiding monetary policy deliberation going forward.

Stability of financial institutions

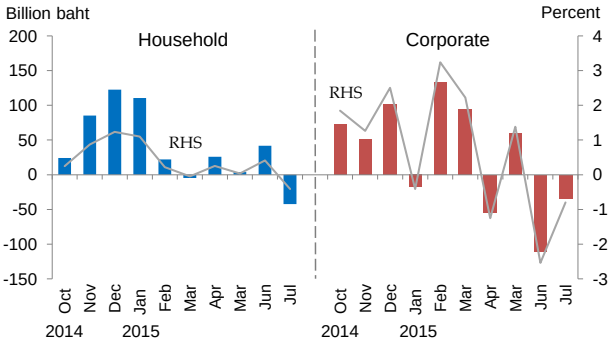
The overall stability of the financial institutions system remains strong, although loan quality deteriorated somewhat following economic conditions. Nevertheless, commercial banks maintained high level of loan loss provision and capital base that serve as cushion against deteriorating loan quality.

Chart 3.15 Special deposit rates*



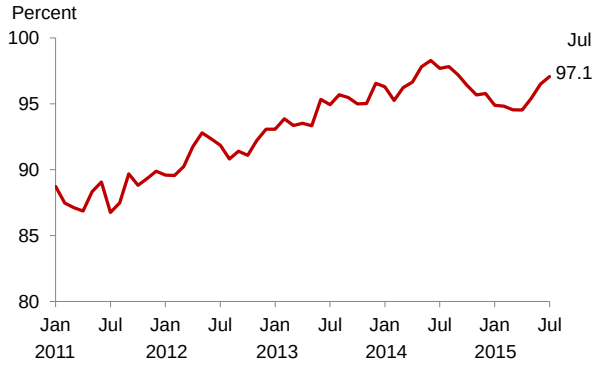
Note: *Maximum rates offered
Source: Bank of Thailand

Chart 3.16 New deposits*



Note: *Change in outstanding deposits at depository institutions (excluding Bank of Thailand). The amounts only account for new deposits, not including transfers within and between commercial banks or rolled-over deposits.

Chart 3.17 Loans to deposits and B/E ratio of commercial banks

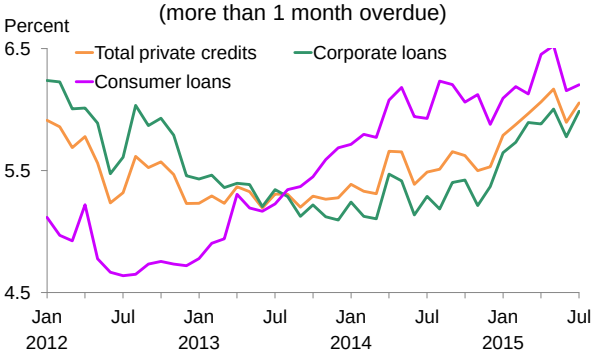


Source: Bank of Thailand

Profitability of commercial banks declined moderately in 2015 Q2, following the economic slowdown. This was indicated by the fall in net profit from 59.8 billion baht at the same period last year to 53.3 billion baht during this period **(Table 3.2)**. The decline was due in part to a growing level of allowance for doubtful accounts, which provided cushion against worsening loan quality. Moreover, net interest income fell in line with slowing credits and falling interest rates. As a result, returns on assets (ROA) decreased from 1.48 percent during the same period last year to 1.26 percent this year.

Weak economic conditions also weighed on repayment ability of the private sector, prompting a decline in business and consumer loan quality, particularly for SME loans, car loans, and credit card loans **(Table 3.2)**. This is evident from the rise of delinquency and NPL ratio in 2015 Q2 **(Chart 3.18)**. However, with prudent risk management and sufficient levels of loan loss provision, commercial banks should be able to withstand falling loan quality. According to the latest data, the actual to regulatory loan loss provision ratio stood at 165.1 percent, while the Capital Adequacy Ratio was at 16.7 percent.

Chart 3.18 Delinquency and NPL Ratio
(more than 1 month overdue)



Note: Excluding loans to government and Interbank loans
Source: Bank of Thailand

3.3 Non-financial sectors

Household sector

Soft economic conditions added risk to the household sector, given the recent downtrends in household incomes and repayment ability. Thus, the impact of economic conditions on these two aspects of households with high debts, especially

those with low income and in the agricultural sector, will require close monitoring going forward.

Subdued economic recovery dampened household incomes, as farm incomes continued to fall both from both quantity and price, and non-farn income (wages including overtime) stabilized **(Chart 3.19)**. As such, households remained cautious in spending and in incuring new debts, causing a fall in household credit growth from 6.6 percent at the end of 2014 to 6.4 percent at the end of 2015 Q1 **(Chart 3.20).**

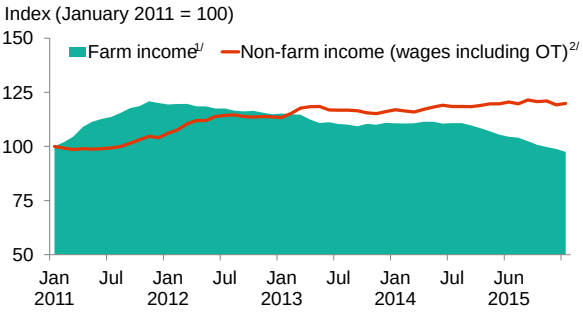
The falling income and high level of outstanding debt reduced household repayment ability, as indicated by the slight increase in the delinquency and NPL ratio from 6.1 percent at the end of Q1 to 6.2 percent at the end of Q2 **(Table 3.2)**, reflected particularly by deterioration in quality of car loans and credit card loans.

Given recent developments, the impact of economic conditions on household income and repayment ability require close monitoring in the future. Special attention should be paid on those with low-income and in agricultural sector affected by the draught, because they possess higher debt service ratio than other groups.

Corporate sector

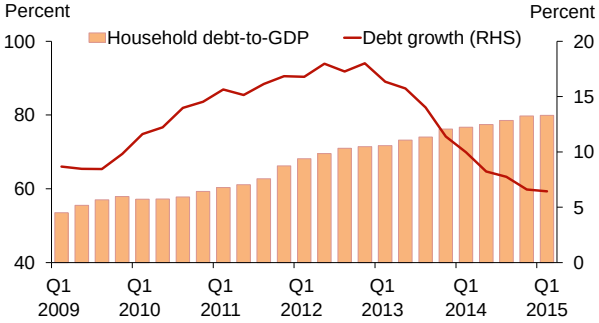
Earnings of listed non-financial companies in 2015 Q2 saw a slight dip from the previous quarter, as a result of slow economic recovery. Although the overall financial standing of corporate sector remained sound, smaller companies experienced weakening repayment ability. In the next period, the impact of economic recovery on corporate earnings and stability of this sector will have to be closly observed.

Chart 3.19 Household income



Note: 1/ Seasonally adjusted, 12-month moving average
2/ Seasonally adjusted, 3-month moving average
Source: Office of Agricultural Economics and National Statistical Office, calculations by the Bank of Thailand

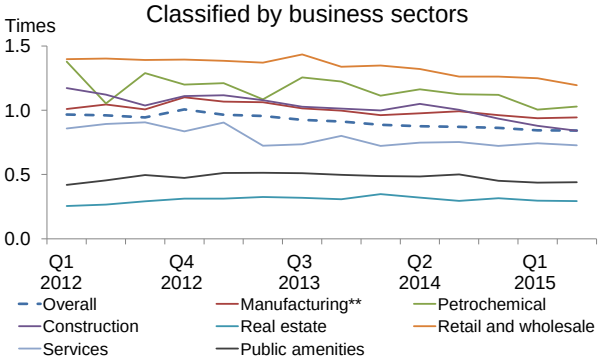
Chart 3.20 Household debt1/ to GDP2/



Note: 1/ Loans to households by financial institutions
2/ Computed using fix-based GDP (base year = 1988)
Source: Bank of Thailand

Current economic slump weighed on earnings of listed non-financial companies, causing sales and profitability to drop from the previous quarter. Asset turnover ratio (ATO) **(Chart 3.21)** edged down, especially in shrinking sales of the wholesale and retail sectors. Depressed sales volumes also lowered profitability, as suggested by the decline in the median of operating profit margin (OPM) from 5 percent in 2015 Q1 to 4.7 percent in Q2 **(Chart 3.22).** Nevertheless, petroleum and petrochemical sectors recorded higher profitability, due to the decline in the cost of crude exceeding the fall in retail prices.

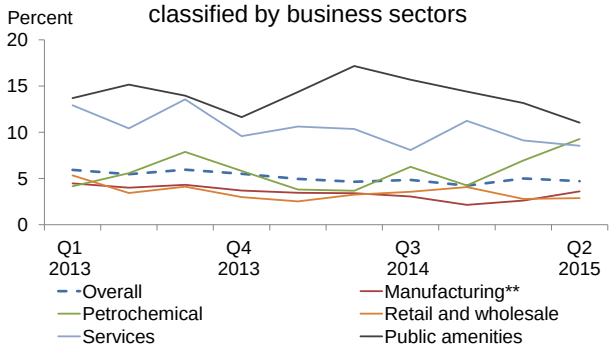
Chart 3.21 Asset Turnover Ratio (ATO)*
Classified by business sectors



Note: *Median values of financial ratios of each sector
**Excluding petrochemical
Source: The Stock Exchange of Thailand. Calculations by the Bank of Thailand

The median annualized interest coverage Ratio (ICR), which indicates corporate repayment ability, stabilized at 4.9 times **(Chart 3.23)**. When listed firms are classified into 5 groups according to their asset sizes, it can be seen that ICR of firms with smallest asset size deteriorated substantially and is set to worsen further. If the condition of these small listed firms is taken to reflect the situations of non-listed SMEs, due to the lack of actual data, this appears to suggest the worsening repayment ability of SMEs as well. The increase in NPL ratio among SMEs is consistent with the observation.

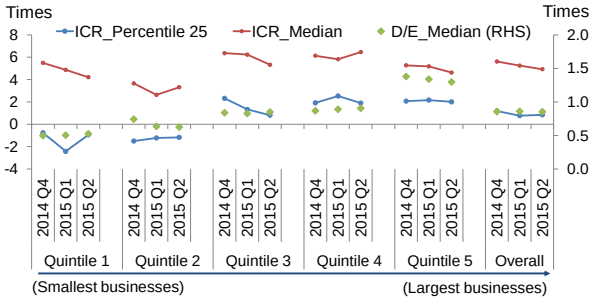
Chart 3.22 Operating Profit Margin (OPM)*
classified by business sectors



Note: *Average financial ratio of each sector
**Excluding petrochemical
Source: The Stock Exchange of Thailand. Calculations by the Bank of Thailand.

Looking ahead, the impact of economic recovery on corporate earnings and financial standing requires close monitoring. Particular attention should be paid to repayment ability of SMEs, which are more fragile due to lower liquidity and tighter credit standards they face.

Chart 3.23 Interest Coverage Ratio (ICR*)
classified by business sizes**



Note: *Annualized data
**Firms are classified into quintiles using asset sizes as of 2015 Q2
Source: The Stock Exchange of Thailand. Calculations by the Bank of Thailand.

Real estate sector

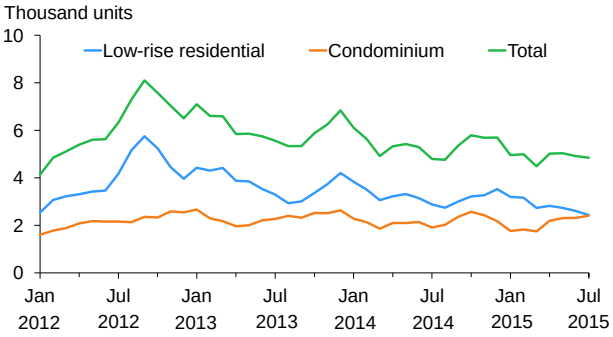
Excess supply in the real estate sector continued to rise, as consumer purchasing power stalled in line with weak economic conditions. Although the overall sector possess solid financial

positions, the development of excess supply should be closely observed in the near future.

Demand in the real estate sector slowed from the previous year, as a result of weak economic recovery, declining household purchasing power, high level of outstanding debts, and tighter home credit standards by financial institutions. The number of new home loans approved by commercial banks over the first 7 months of this year dropped by 7 percent from the same period last year **(Chart 3.24)**. Most of the fall was accounted for by low-rise projects, while new credits for condominium units expanded somewhat. Meanwhile, real estate firms continued to launch new projects, especially condominiums **(Chart 3.25)**, partly in anticipation of the extensions of mass transit rail routes. With demand lagging behind supply, excess supply has continued to climb. This is shown by the number of cumulative unsold housing units in Bangkok and vicinity over the first 7 months of 2015, which was 13 percent larger than that of the same period last year **(Chart 3.26)**. Since the majority of excess supply was in condominium units priced below 2 million baht, it reflects the weakening confidence and purchasing power among low to middle income households.

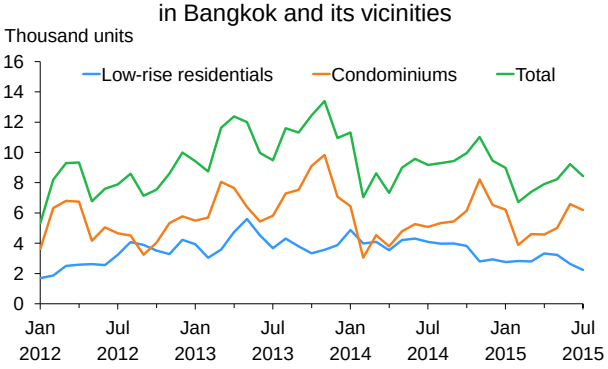
As for the financial position of real estate companies, most developers have maintained solid financial standing. As a result, they should be able to withstand the financial challenges associated with an increase in excess supply to a certain extent. The median asset turnover ratio (ATO) and current ratio (CR) of the sector stood at 0.3 and 2.4 times, respectively, close to the 5-year past averages. Yet, loan repayment ability of some firms started to deteriorate, as reflected by the fall in the 25th percentile of interest coverage

Chart 3.24 New residential units with loans from commercial banks in Bangkok and vicinity



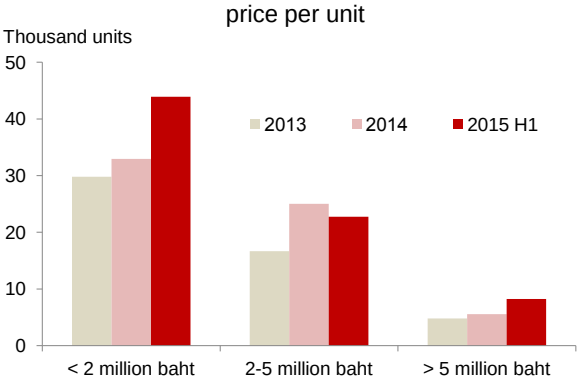
Note: *3-month moving average
Source: Agency for Real Estate Affairs (AREA) and calculations by Bank of Thailand

Chart 3.25 New residential project launches in Bangkok and its vicinities



Note: *3-month moving average
Source: Agency for Real Estate Affairs (AREA) and calculations by Bank of Thailand

Chart 3.26 Unsold condominiums classified by price per unit

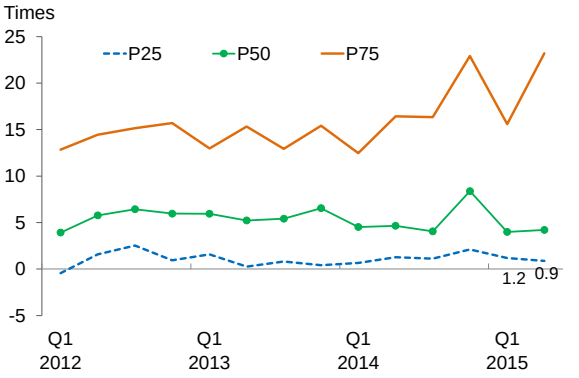


Source: Agency for Real Estate Affairs (AREA) and calculations by Bank of Thailand

ratio (ICR) from 1.2 times in the first quarter to 0.9 times in 2015 Q2 **(Chart 3.27)**. Moreover, the pre-finance loan quality worsened slightly **(Chart 3.28)**, while NPL ratio stood at 4.4 percent in 2015 Q2, up from 4.3 percent in the previous quarter.

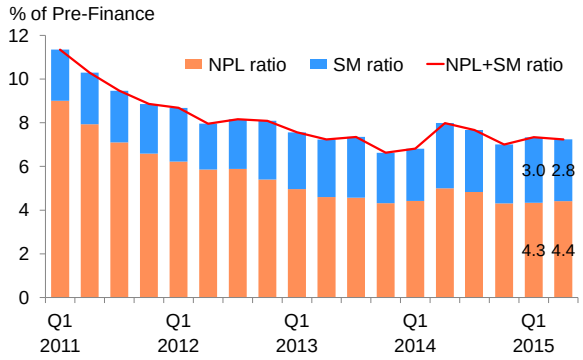
Going forward, stability of the real estate sector will depend on clarity of the economic recovery influencing home purchasing decisions, as well as the ability of real estate companies to respond to growing excess supply.

Chart 3.27 ICR of real estate companies



Source: The Stock Exchange of Thailand. Calculations by the Bank of Thailand.

Chart 3.28 Quality of loans to real estate companies in the commercial banks system (Pre-Finance)



Source: Bank of Thailand

Fiscal sector

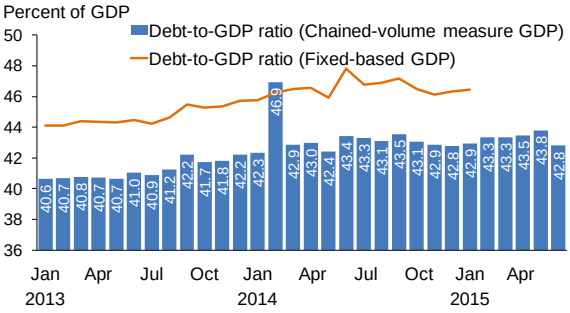
Stability of the fiscal sector continued to be robust overall. Treasury cash balance remained adequate to buffer the lower-than-projected tax revenue collection. Moreover, the level of public debt remained low, a satisfactory level. In the future, although public debt is set to rise due to additional borrowings needed to cover budget deficit in the 2016 fiscal year, investment in two new motorway routes, and investment in infrastructures according to the Action Plan, these debts should not give rise to concerns, for they will be for raising the country's economic potentials. Meanwhile, quasi-fiscal activities aimed at stimulating the economy will require further attention, for they may raise future level of public debt.

The outlook for fiscal sector remained satisfactory, as reflected by a number of indicators. (1) Treasury budget balance had increased and remained sufficient to meet necessary demand of liquidity as well as offset the projected shortfall of revenue collection this year, marking two consecutive years of revenue shortfall in line with economic slowdown. Treasury budget balance at the end of June 2015 stood at 264 billion baht, up from 126 billion baht in April 2015. The increase was a

result of the delay in collection of a portion of corporate tax from May to June[4/]. Moreover, there was an increase in tax receipts from revenues of state enterprises and contribution from excess liquidity of revolving funds. **(2) The ratio of public debt to GDP fell slightly, and remained well below the fiscal sustainability threshold of 60 percent.** In June, 2015, the ratio recorded at 42.8 percent, down from 43.5 percent in April 2015 **(Chart 3.29)**. The redemption of matured government and corporate bonds, together with capital repayments by financial and non-financial state enterprises, contributed to the fall in the ratio.

Going forward, public debt is expected to grow from borrowings required to cover budget deficits in 2016 and finance two new motorway routes, and also to invest in infrastructures according to the transportation Action Plan of 2015-2016. However, the rising debt should not give rise to concerns, since it will contribute to the increase of the country's economic potentials. Nevertheless, quasi-fiscal activities aimed at stimulating the economy will require close attention, for they may push level of public debt higher in the future.

Chart 3.29 Public Debt to GDP ratio*



Note: (1) Chart shows calendar years, (2) Official figures for Debt-to-GDP ratio based on Chained-Volume Measure GDP are only available for February and March 2015. The figures prior to this period are calculated by the Bank of Thailand.

Source: Public Debt Management Office

[4/] The original deadline for corporate tax payment (May 30, 2015), fell on a public holiday, so it was postponed to June 2, 2015. Moreover, due to the growing amount of online payment, the deadline was extended by 8 days.

Table 3.2 Sectoral Indicators for assessing risks and vulnerabilities to financial stability

Indicators	2014	2014			2015			
		H1	Q3	Q4	Q1	Q2	Jul	Aug
1. Financial markets sector								
Bond market								
Bond spread (10 years-2 years)	1.3	1.4	1.2	1.4	1.4	1.4	1.3	1.2
Equity market								
SET Index (End of period)	1,497.67	1,485.80	1,585.70	1,497.70	1,505.94	1,504.55	1,440.12	1,382.41
Actual volatility (SET Index)[1/]	12.51	14.14	8.35	13.43	12.83	11.29	12.54	24.57
Price to Earnings Ratio (times)	17.8	17.9	18.5	17.8	20.9	20.1	19.2	18.2
FX market								
Actual volatility (baht) (%annualized)[2/]	4.0	4.3	3.8	3.4	3.7	5.4	4.4	5.3
Nominal Effective Exchange Rate (NEER)	104.3	102.7	104.6	107	112	110	108	106
Real Effective Exchange Rate (REER)	103	101.9	103.4	105	108	105.9	104	n.a
2. Financial institutions sector[3/]								
Minimum lending rate (MLR)[4/]	6.8	6.8	6.8	6.8	6.6	6.5	6.5	6.5
12-month fixed deposit rate[4/]	1.7	1.7	1.7	1.7	1.5	1.5	1.5	1.5
Capital adequacy								
Regulatory capital to risk-weighted asset (%)	16.8	15.9	17.1	16.8	16.6	16.7	16.7	n.a.
Earnings and profitability								
Net profit (billion baht)	214.2	59.8	53.8	48.7	52.5	53.3		
Return on assets (ROA)	1.32	1.48	1.33	1.17	1.21	1.26		
Liquidity								
Loan to deposit and B/E	95.7	98.1	97.2	95.7	94.5	96.5	97.1	n.a.
3. Household sector								
Household debt to GDP (times)	79.7	77.4	78.5	79.7	79.9	n.a.		
Financial assets to debt (times)	2.1	2.0	2.0	2.1	n.a.	n.a.		
NPL and delinquency ratio (%)								
Thai commercial banks :								
Consumer loans	5.9	6.0	6.2	5.9	6.1	6.2		
Housing loans	3.8	3.8	4.1	3.8	4.1	4.0		
Auto leasing	10.8	10.5	10.6	10.8	10.9	11.1		
Credit cards	5.3	5.5	6.1	5.3	6.3	6.8		
Other personal loans	5.0	5.0	5.6	5.0	5.4	5.3		
4. Non-financial corporate sector [5/]								
Operating profit margin (%)	4.7	4.8	4.8	4.2	5.0	4.7		
Debt to equity ratio (times)	0.8	0.8	0.8	0.8	0.7	0.7		
Income coverage ratio (times)	6.0	6.2	5.9	5.6	5.3	5.3		
Current ratio (times)	1.6	1.6	1.6	1.6	1.7	1.6		

Table 3.2 Sectoral Indicators for assessing risks and vulnerabilities to financial stability

Indicators	2014	2014			2015			
		H1	Q3	Q4	Q1	Q2	Jul	Aug
4. Non-financial corporate sector [5/] (Continues)								
Delinquency and NPL ratio								
Large bussiness	1.1	1.2	1.1	1.1	1.1	1.2		
SMEs	3.1	3.4	3.4	3.1	3.3	3.4		
5. Real estate sector								
The number of approved mortgages from banks (Bangkok and its vicinity)	62,839	16,315	17,345	16,299	11,564	14,939	5,044	
Single-detached and semi-detached houses	15,694	4,206	4,230	3,927	3,001	3,383	901	
Townhouses and commercial buildings	21,764	5,921	5,844	5,215	4,212	4,987	1,404	
Condominiums	25,381	6,188	7,271	7,157	4,351	6,569	2,739	
The number of new openings (Bangkok and its vicinity)	111,211	28,714	28,268	28,367	22,191	27,677	5,914	
Single-detached and semi-detached houses	18,933	5,657	5,349	3,173	3,231	2,045	2,055	
Townhouses and commercial buildings	26,980	7,261	6,611	5,609	5,172	5,866	1,156	
Condominiums	65,298	15,796	16,308	19,585	13,788	19,766	3,859	
Housing price index [6/]								
Single-detached houses (including land)	117	115.5	119.3	118.4	119.4	118.4	119.6	
Townhouses (including land)	130.2	128.6	134.2	132.5	135.8	138.2	139.8	
Condominiums	154.3	153.1	156	161.9	162.9	164.4	164.6	
Land	150.7	148.3	153	155.5	163.1	163.3	164.5	
6. Fiscal sector								
Public debt to GDP (%)	46.3	47.8	47.2	46.3	43.3	42.8		

[1/] Annualized standard deviation of return

[2/] Daily volatility (using exponentially weighted moving average method)

[3/] Based on data of all commercial banks

[4/] Average value of 4 largest Thai commercial banks

[5/] Only listed companies on SET (median)

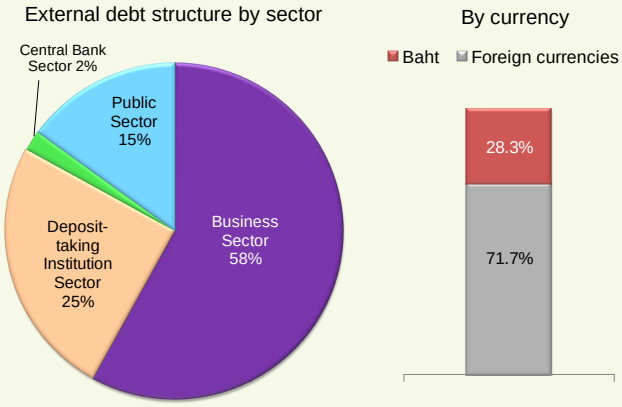
[6/] Due to the fact that the structure of the housing market has changed significantly, the Bank of Thailand is currently improving the price index to better reflect the structure change

Baht depreciation: implications on external debt and external stability

Over the past few months, the Thai baht and emerging market currencies persistently depreciated due to the expected tightening of the Fed's policy rate later this year, the economic slowdown in China and other Asian economies, as well as the renminbi devaluation. While the weak baht may bolster the recovery in the exports sector, the inevitable question is whether the increased external debt burden in baht term leading to external vulnerability similar to the 1997 financial crisis. This box is an assessment of **the impacts of the weakening baht on Thailand's external debt burden and external stability.**

Assessment of Thailand's external debt burden risk in the second quarter of 2015 finds that Thailand's external debt stood at 136 billion U.S. dollars, accounting for 33.3 percent of Thailand's GDP, which remains well below the prudential limit according to international standards[1/]. Moreover, a closer look into the country's external debt structure and the practice of borrowers with foreign debt finds that **the impact of the baht depreciation on external debt burden remains limited** due to the following reasons:

Chart 1 Thailand's external debt structure by sector and by currency as of 2015 Q2



Source: Bank of Thailand

(1) Ratio of baht-denominated external Debt

At present, 28 percent of Thailand's external debt is denominated in baht **(Chart 1)**, unlike in 1997 when nearly all of the external debt was denominated in foreign currencies. The baht-denominated debt is in the forms of government bonds and Bank of Thailand bonds held by foreign investors, and some private sector baht-denominated loans, especially those belonging to holding companies. **This baht-denominated debt burden is unaffected by the baht depreciation.** Although the weakening baht may cause some foreign investors to offload their baht-denominated bonds, which may further exert weakening pressure on the baht, this would not impact the debt burden itself.

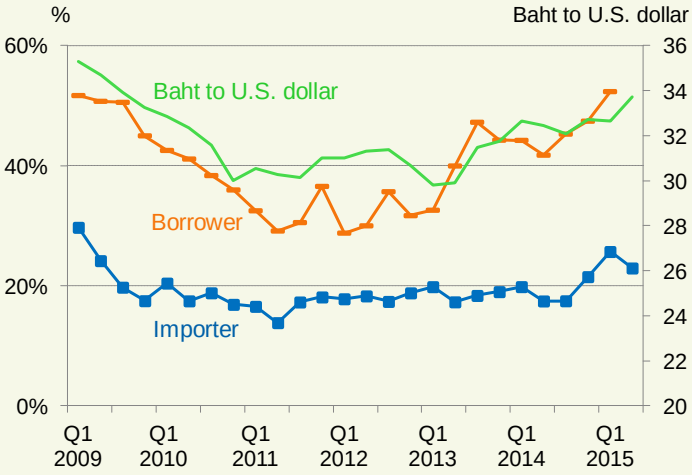
(2) Currency risk management practice by the private sector remains sound

Foreign-currency-denominated debt, which makes up 72 percent of total external debt, **are mostly hedged against risks from exchange rate fluctuations**, due to the lesson learned from the 1997 economic crisis. The impact of the baht depreciation on foreign-currency-denominated debt is therefore limited. Within **non-financial sector**, a vast majority of the foreign currency debt is held by large businesses. An analysis of the top 50 holders of foreign currency debt (as of 2014 Q4), which contribute to approximately 50 percent of total external debt held by corporate sector, finds that **most of these business have engaged in**

[1/] The prudent external debt-to-GDP ratio for middle income countries is between 48 – 80 percent according to the World Bank.

proper hedging activities. Some of the businesses receive export earnings and returns on investment in foreign currencies, which act as a natural hedge for their foreign currency-denominated debt. These are businesses in sectors such as energy, electricity, automobile, tyres and food. Meanwhile, businesses whose revenues are in baht, normally hedge against currency risks. For example, while the non-deposit-taking leasing companies borrow in foreign currencies and lend in baht, they fully or largely hedge against exchange rate risks. The value of the debt burden in baht term is therefore relatively unaffected from exchange rate movement. Meanwhile, major net importers, such as the petroleum industry, are able to pass on a large proportion of the increased cost resulting from baht depreciation towards the domestic retail prices, while benefiting from increased export incomes. In addition, in response to the baht depreciation over the recent months, **importers and foreign currency borrowers have taken further cushion against exchange rate risk by increasing their own hedging ratios**. This reflects the close monitoring and prudent practices of the private sector under the changing environment **(Chart 2)**.

Chart 2 Hedging ratio



Note: 1) Borrower's hedging ratio is calculated from the outstanding amount in the foreign currency forward contracts held by commercial banks' customers for the repayment of both short-term debts and long-term debts with no more than 1 year maturity.
2) Importer's hedging ratio is calculated from the outstanding amount in importer's foreign currency forward contracts to the importer's 3-month import value

Source: Bank of Thailand

Risks arising from foreign-currency-denominated debts of **financial institutions** also remain limited. This can be attributed to the Bank of Thailand regulations, which mandates commercial banks to limit their foreign currency-denominated net assets and liabilities. At the same time, commercial banks engage in proper risk management measures, for example, by ensuring overall currency and maturity matches for both assets-and liabilities. Commercial banks' short-term external debts are taken on to square positions as they provided hedging transactions for the business sector. Long-term external debts are mostly incurred to facilitate outward direct investments, which have been on an uptrend since 2012, by Thai companies in other countries. The current situation differs from the 1997 financial crisis when most of the country's external debts were short-term debts taken on for long-term investments in businesses whose earnings were not in foreign currencies, such as in the real estate sector. These currency and maturity mismatches were the main sources of external risks to financial stability during the 1997 crisis.

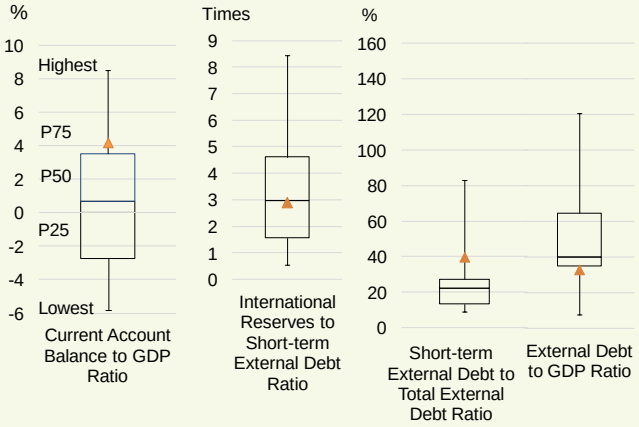
Assessment of Thailand's External Stability

Robust external stability enables the economy to be able to effectively counter volatilities in the global financial markets. It also reduces the risks of liquidity shortage or missed foreign debt payments, which may lead to the balance of payments crises. When assessing external stability of an economy, factors considered must include the flexibility of

the economy's financial positions in absorbing risks and the public sector's ability to implement policies in response to such risks. Under this framework, **Thailand's external stability is assessed to be sound and robust.**

(1) Flexibility of the economy in absorbing risks depends on the economy's reliance on foreign capital, as well as its liquidity position and external debt burden. A comparison of key external stability indicators among emerging market economies[2/] can be found in **Chart 3**. The indicators include:

Chart 3 Thailand's key external stability indicators, in comparison with other emerging market economies



Note: 1) P25, P50, and P75 indicate the 25th, 50th, and 75th percentile of the 22 emerging economies, respectively
2) International benchmark:
Current Account Balance to GDP Ratio > -2 percent
International Reserves to Short-term External Debt Ratio > 1 time
External Debt to GDP Ratio < 48 percent

Source: BOT, IMF, IIF and World Bank

◆ Current Account Balance to GDP Ratio

Over the recent quarters, **the Thai economy has continued to run a current account surplus**, with an expected increase in surplus in 2015. This is due, in part, to lower imports driven by the economic slowdown, and a plunge in the value of oil imports as a result of the sharp decline in the global crude oil price. In this respect, Thailand differs from the oil exporting countries whose current account balances run a shrinking surplus or growing deficit. The Bank of Thailand expects that Thailand's current account surplus this year is likely to stand at 25.5 billion U.S. dollar. Therefore this reflects Thailand's ample excess liquidity from domestic savings to finance the country's investment, which reduces the country's vulnerability in the event of sharp capital outflows. As of 2015 Q2, Thailand's current account balance as a percentage of GDP stood at 4.2 percent – well above the median of 0.7 percent for the emerging market economies.

◆ External Debt to GDP Ratio

A country's external debt to GDP ratio reflects its debt repayment ability. **For the case of Thailand, the ratio stands at 33.3 percent (as of 2015 Q2), indicating Thailand's dependence on external borrowing is below the median of for the emerging markets** at 40 percent. Moreover, the country has strong debt repayment ability, as reflected in the debt service ratio of 4.9 percent – well below the internationally accepted threshold of 20 percent according to the World Bank.

◆ Liquidity indicators, including short-term external debt to total external debt ratio and international reserves to short-term external debt ratio

The short-term external debt to total external debt ratio reflects the level of dependency on short-term external debt. A high ratio indicates that a country runs a high risk

[2/] The 22 emerging markets include Argentina, Brazil, Bulgaria, Chile, China, Colombia, Czech Republic, Hungary, India, Indonesia, the Republic of Korea, Malaysia, Mexico, Peru, the Philippines, Poland, Romania, Russia, South Africa, Thailand, Turkey and Ukraine.

in the event of foreign currency liquidity shortage. For Thailand, the ratio currently stands at 40 percent, higher than the median of the emerging markets. In the broad picture, however, **such figure does not reflect a worrying level of risks to Thailand's external liquidity position,** as the majority of Thailand's short-term debt is foreign currencies borrowed by commercial banks for currency hedging transactions demanded by the business sector. Hedging transactions only possess limited currency and maturity mismatches. Most importantly, the amount of short-term debt is relatively small in comparison to international reserves, as indicated by the **ratio of international reserves to short-term external debt**. Analysts and investors attach high importance to this ratio in assessing a country's risks to the balance of payments crisis. This indicator reflects the country's ability to absorb shocks from potential capital flow volatiles. At present, **the ratio currently stands at 2.9 times for Thailand**, well above the international threshold of one-to-one. Thailand, therefore, maintains ample level of international reserves for the repayment of its short-term debt in case the existing short-term debt cannot be extended.

(2) The government's ability to implement policies in response to risks

The effectiveness of maintaining external stability and balance also depends on the policy space available to policy-makers, on both monetary and fiscal sides.

◆ Monetary Policy

While inflationary pressures are low and the risk to financial stability is limited, monetary policy stance can continue to be sufficiently accommodative to lend support to economic recovery. Looking ahead, should any factors pose a risk to Thailand's external stability, **the Bank of Thailand has adequate policy tools to counter those risks.** In addition, under the inflation targeting framework and the managed float regime, **the exchange rate is allowed to adjust to market mechanisms and economic fundamentals, reducing the pre-conditions for currency speculative attacks.** The Bank of Thailand will intervene in cases of sudden and extreme movements of the exchange rate to the extent that it would negatively affect the confidence and the adjustment ability of the private sector. Moreover, the current amount of international reserves is sufficient to respond to volatilities in the global financial markets.

◆ Fiscal Policy

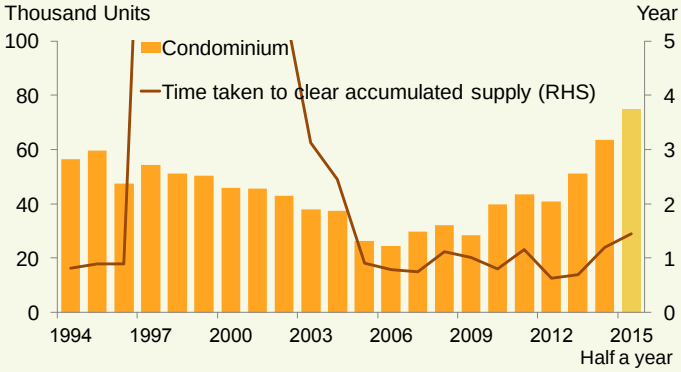
Thailand has sufficient fiscal policy space to maintain macroeconomic stability resulting from volatile capital flows, as reflected by its low fiscal deficit to GDP ratio of -1.9 percent in the 2015 fiscal year, and its public debt to GDP ratio of 42.8 percent (as of July 2015), which is well below the 60 percent limit of the fiscal sustainability framework. Thailand therefore has the policy flexibility to respond to changing external environment, if necessary.

In conclusion, the MPC judged that a sharp depreciation of the baht will likely have limited impacts on Thailand's external stability. However, against the backdrop of the uncertainties in the global financial markets and weak domestic economic recovery, there is a possibility that baht volatility will persist, which may potentially impact the country's economic stability in the periods ahead. **Therefore, the public sector and the private sector should collaborate in monitoring and assessing various risk factors as well as undertake necessary adjustments and appropriate risk management measures in a proper and timely manner.**

Assessing the Risks of Cumulative Excess Supply and Speculation in the Real Estate Market

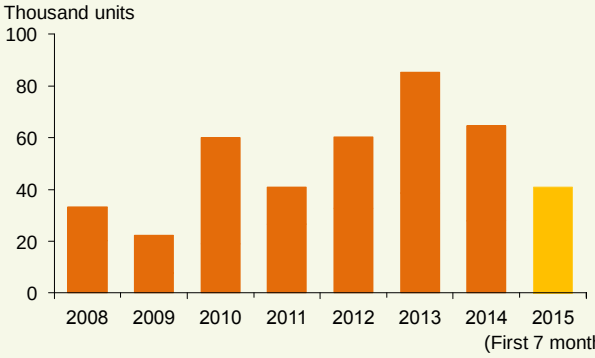
Over the past 3 – 4 years, the cumulative excess supply of housing, especially condominiums in the Bangkok metropolitan area, has been steadily increasing[1/] **(Chart 1).** This is a result of waning demand in the real estate market driven by the slow economic recovery, while the real estate developers continued to launch new projects. **Cumulative supply of condominium units is expected to continue to build up** for the second half of 2015 on account of the following factors. (1) Several condominium projects are to be completed within this year, as many projects were launched 2-3 years ago. (2) Developers also plan to launch a large number of new projects this year **(Chart 2)**. (3) The proportion of the transfer of ownership to the number of units reserved two years prior has decreased since 2014 Q3 **(Chart 3)**, partly attributable to the delay in the construction progress, but this could be due to the buyers changed their minds about owning the properties or failure to obtain mortgage from financial institutions to complete the transfers. There is therefore high likelihood that cumulative excess supply of condominiums will continue to rise, going forward.

Chart 1 Accumulated supply in condominium buildings in Bangkok Metropolitan Region at year's end, by type



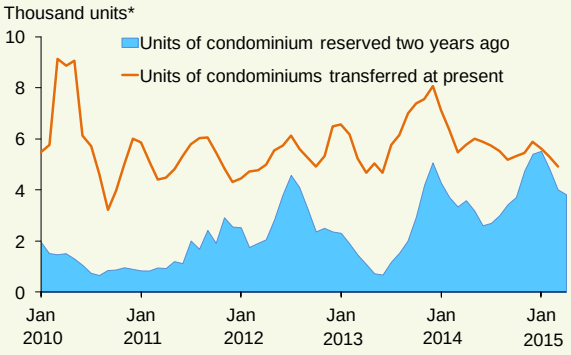
Note: Time taken to clear accumulated supply of condominium is calculated using the number of accumulated supply divided by the number of condominium units sold in the same year.
Source: Agency for Real Estate Affairs, calculations by the Bank of Thailand

Chart 2 Number of new condominium projects launched in Bangkok Metropolitan Area



Source: Agency for Real Estate Affairs (AREA), calculations by the Bank of Thailand

Chart 3 Condominium unit reservations and transfer of ownership

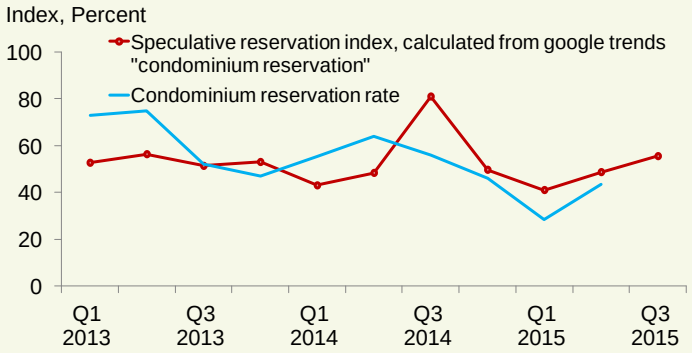


Note: * 3-month moving average
Source: Agency for Real Estate Affairs (AREA) and Department of Lands, calculations by the BOT

[1/] While the number of excess supply of condominium unit is currently higher than in 1997, there is a smaller implication on the excess supply risks because the size of the economy has grown by nearly 3 folds, and there is now a stronger preference in buying condominium units for living. Therefore, the demand should be able to better absorb the supply. It is expected that the excess supply in condominium units can be absorbed within one and a half years, whereas it would have taken more than 10 years in 1997.

Moreover, the short-term speculation through the buying and reselling of property reservation has gone up since the first quarter of the year, as reflected by the speculative reservation index[2/] **(Chart 4)**. Such trend is consistent with the information received from real estate developers that they have seen an increase in short-term speculation through trading of reservations. If the increase in the speculation transactions exceed the developers' estimate, it may mislead developers into launching new condominium projects based on inaccurate information. In the event that speculators are unable to find real buyers for these condominium units once the construction is completed, these units then lead to additional financial burden for developers and worsen the excess supply in the market.

Chart 4 Condominium reservation rate and speculative reservation index for Bangkok Metropolitan Area



Source: Condominium reservation rate from AREA, calculated from the number of condominium units reserved during the first month of the launch to the number of new units launched in the same month, with quarterly calculations by the BOT. Speculative reservation index from google.com/trends, using search keyword "reservation condominium", calculations by the BOT.

Assessments of Real Estate Sector Risks to Financial Stability

Despite the increasing risk from excess supply and short-term speculations in the condominium market, **its impacts at present on financial and macroeconomic stability remains limited.** This is because the majority of the market belongs to large listed companies with sound risk management measures and strong financial position, which enable them to cushion potential financial impacts. The developers have also adjusted to the changing market environment, including focusing its projects towards buyers with higher purchasing powers and delaying or cancelling projects with low reservation rate.

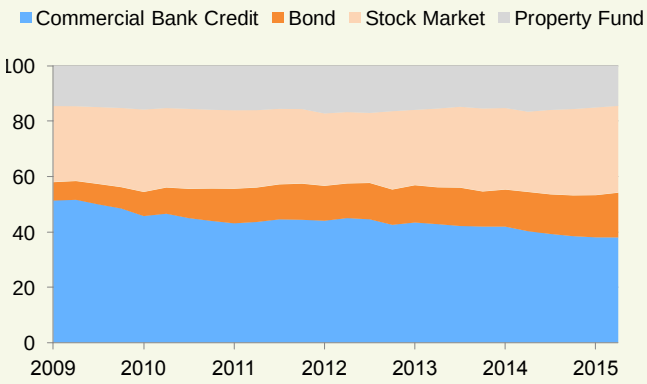
In addition, over recent months, commercial banks have become more cautious in extending loans, both in the pre-financing process for developers, especially through close monitoring of speculative reservations, and in the post-financing process through careful evaluation of the individuals' debt repayment ability. Nonetheless, commercial banks have not been too stringent to the point where creditworthy borrowers are unable to borrow, as reflected by the ratio of housing price to borrower's income, which remains close to the past averages . Such careful monitoring and solid financial positions of commercial banks play an important role in reducing the risks to financial stability and in mitigating the impacts from excess supply and speculation in the real estate sector

However, there is also a trend among major developers to be less dependent on commercial bank loans. Instead, companies look for capital from other channels, such as

[2/] Information gathered from reservation re-sell announcements and condominium reservation information found online through Google.

bonds[3/], joint ventures or property funds **(Chart 5)**. Such shift shows positive development in the capital market, which becomes an increasingly important source of finance, in addition to commercial banks. That said, it, however, creates a channel where risks from the real estate sector can directly impact the capital market, affecting institutional and individual investors alike. The MPC therefore continues to monitor the economic recovery, influencing property demands. The Committee also pays attention to the adjustments made by developers in response to the growing excess supply in the real estate market. If the economic recovery continued to be slow, there is a risk that demand for real estate would also further deteriorate, leading to price reduction for condominium units. This would undoubtedly weigh on the income and debt repayment ability of real estate developers, as well as posing greater risks to financial stability in the future.

Chart 5 Real estate developers' sources of capital



Source: SEC, SET, BOT

Note: "Property Fund" includes both property fund and REITs

[3/] During the first 7 months of 2015, there have been approximately 1,200 billion baht worth of bonds issued by real estate developers. Most of these bonds have been rated by credit rating agencies. The non-rated bonds in the real estate sector constitute 5 percent of the total new bond issuance.